Salivary gland organoids: from biological complexity to precision medicine and regeneration

Yeo-Jun Yoon[1], Ye Jin Jeong[1], Minjeong Jo[1], Seungyeon Hwang[1], Sunyoung Seo[1], Hyewon Jeong[1], Toan V. Phan[2,3], João N. Ferreira[4], and Jae-Yol Lim[1,5,*]

Running title: Salivary Organoids: From Precision to Regeneration

[1]Department of Otorhinolaryngology, Yonsei University College of Medicine, Seoul, 03722, Republic of Korea

[2]Department of Oral Pathology and Maxillofacial Surgery, Faculty of Dentistry, Van Lang University, Ho Chi Minh City, Vietnam

[3]Graduate Program in Oral and Craniofacial Science, Faculty of Dentistry, Chulalongkorn University, Bangkok, Thailand

[4]Avatar Biotechnologies for Oral Health and Healthy Longevity Research Unit, Department of Research Affairs, Faculty of Dentistry, Chulalongkorn University, Bangkok, 10330, Thailand

[5]Department of Otorhinolaryngology, Gangnam Severance Hospital, Yonsei University College of Medicine, Seoul, 06273, Republic of Korea

E-mail: yeojuny@yuhs.ac, yejin3255@yuhs.ac, gratiaa821@yuhs.ac, tmddus110523@yuhs.ac, gagseo@yuhs.ac, 6678303832@student.chula.ac.th, Joao.F@chula.ac.th

[*]**Corresponding authors:**

Jae-Yol Lim, MD, Ph.D., Department of Otorhinolaryngology, Gangnam Severance Hospital, Yonsei University College of Medicine, 211 Eonju-ro, Gangnam-gu, Seoul 06273, Republic of Korea

E-mail: jylimmd@yuhs.ac; Tel.: 82-2-2019-3460; Fax: 82-2-3463-4750

Abstract

Salivary gland disorders span a pathological spectrum, from histologically diverse malignancies to autoimmune epithelial dysfunction in Sjögren's disease and poorly reversible radiation-induced hypofunction, each representing a major unmet clinical need inadequately modeled by conventional preclinical systems. Self-organizing salivary gland organoids offer a physiologically relevant alternative. We first consider organoid generation from adult epithelial and pluripotent stem cells, the WNT–R-spondin, fibroblast growth factor and neuregulin signals sustaining long-term human culture, and assays validating structural and secretory function. A recurring constraint is that human organoids remain ductal-dominant with incomplete acinar maturation, although murine differentiation also varies across protocols; we assess strategies aimed at improving functional maturity. We then examine disease modeling, including patient-derived tumor organoids for subtype-specific drug screening and models of Sjögren's disease and radiation injury, while distinguishing established epithelial findings from prospective immune– and stromal–epithelial systems. We evaluate organoids as quantitative pharmacology platforms for botulinum neurotoxin potency testing and secretagogue screening. Finally, we appraise regenerative strategies encompassing organoid-derived cell transplantation entering first-in-human evaluation, chemically reprogrammed epithelial progenitors and targeted extracellular vesicles, as well as manufacturing, standardization and regulatory barriers to clinical use.

Keywords: human salivary gland organoids, species differences, acinar maturation, patient-derived organoids, precision medicine, immune–epithelial crosstalk, radiation injury, chemical reprogramming, regenerative medicine, standardization

INTRODUCTION

The salivary glands sustain oral health by secreting 0.5–1.5 L of saliva each day through a branched architecture of secretory acini and a ductal network that refines the ionic composition of the primary fluid (Fig. 1)[1]. This output depends on the coordinated activity of serous and mucous acinar cells, intercalated and striated duct cells, myoepithelial cells and a dense neurovascular supply, and this interdependence becomes clinically consequential when it fails[2-4]. Two disease burdens dominate. Radiotherapy for head and neck cancer, autoimmune destruction in Sjögren's disease and age-related atrophy converge on irreversible hypofunction and xerostomia, for which treatment remains palliative[5,6]. Salivary gland malignancies, by contrast, display the widest histological diversity of any organ, with more than twenty World Health Organization (WHO)-recognized subtypes, each demanding individualized management[7]. Underlying both is a research problem: the available models capture little of this biology. Two-dimensional (2D) cultures lose glandular architecture, and available cell lines and animal models incompletely capture disease heterogeneity and the therapeutic responses[8,9].

Organoids offer a way past these limits. As self-organizing three-dimensional (3D) cultures that reconstruct tissue architecture, cellular composition and physiological function, they have opened salivary gland development, disease and regeneration to experimental access in a human-relevant setting[10-12]. The field has matured quickly. Long-term culture systems established from adult murine and human salivary tissues[13-15] defined key niche requirements and species-dependent properties; patient-derived tumor libraries[16,17], stromal- and endothelial-integrated developmental models[18,19], patient-derived Sjögren's disease organoids[20], and progenitor-cell transplantation followed. That trajectory now spans preclinical transplantation, chemically defined human organoids, clinical-grade manufacturing, and clinically annotated biorepositories[15,21-23], marking a transition from proof of concept toward clinical application. The underlying evidence nevertheless spans murine developmental and preclinical models,

human organoid cultures, and transplantation of human-derived cells into mouse recipients, each with distinct implications for clinical translation.

Existing reviews have largely viewed salivary gland organoids through a single lens, whether as regenerative constructs for xerostomia, as substrates for bioengineering and bioprinting, or as one tissue within broader surveys of oral and dental organoids[24-28]. What has been missing is an account that treats the organoid at once as a precision-oncology platform, an autoimmune disease model, a quantitative pharmacology tool and a regenerative therapy, and that confronts the constraint common to all four: human organoids often remain ductal-dominant and show incomplete, protocol-dependent acinar maturation. This review provides that synthesis. Because murine and human organoids diverge in ways that bear on every application that follows, we annotate the species origin of key evidence throughout and consolidate the detailed comparison in a dedicated section (Species Differences: Murine versus Human Organoid Models).

We organize it around three unresolved tensions. The first is developmental and concerns why human organoids stall short of functional acinar maturation and whether niche engineering and defined differentiation can close the gap. The second is architectural. Most current systems are epithelial-only, omitting the stromal, immune and matrix compartments that shape malignancy and Sjögren's disease and so limiting their fidelity as disease models. The third is translational. Transplantable cell products, chimeric models and pharmacology assays each demand manufacturing and regulatory frameworks the field is only beginning to build. Across the sections that follow, we trace these tensions from organoid generation and functional validation, through tumor and autoimmune disease modeling and pharmacological application, to regenerative therapy and the standardization challenges that will decide clinical impact.

PRINCIPLES OF ORGANOID GENERATION

Cell Sources

The starting cells shape what a salivary gland organoid can model, how reproducibly it forms, and whether it can reach the clinic[11]. Three sources dominate, each suited to different questions. Adult epithelial cells are the most accessible. Unlike the intestine, where LGR5 lineage tracing defines a clear stem-cell hierarchy[29], the salivary gland does not conform to a comparably simple model; progenitor populations can generate both ductal and acinar descendants[30] and show broader plasticity after severe injury or removal from the native niche[31]. Establishment and differentiation are therefore less predictable than in organs with defined stem compartments[9]. Their compensating strengths are an intrinsic bias toward salivary lineages[13] and direct suitability for autologous therapy[15], though limited proliferative capacity and age-related decline constrain both[32].

Pluripotent stem cells (PSCs), particularly induced pluripotent stem cells (iPSCs), trade that lineage bias for scale and reach. Derived from accessible somatic cells and effectively unlimited in expansion[33,34], they enable disease modeling that primary tissue cannot support, particularly for genetic conditions and rare disorders where sampling is difficult. In Sjögren's disease, where diagnostic biopsies typically sample labial minor glands, iPSC-based approaches could provide a route to modeling major-gland lineages without direct major-gland sampling, although this remains a prospective application[35]. They have been used to model salivary viral tropism[36] and hold promise for autoimmune systems[35]. PSC-based salivary-gland generation uses distinct strategies, including directed differentiation toward salivary-gland lineages from human iPSCs[37] and conditional blastocyst complementation in mice[38]. The capacity of PSC-derived salivary models to reproduce mature adult gland function remains approach-dependent and requires direct functional validation [36–38,47,35,39], while their infrastructure and extended timelines can complicate applications that need rapid

turnaround39,[40].

Embryonic and fetal tissues serve as the developmental reference against which the other sources are judged. Rich in highly proliferative progenitors that adults cannot supply [41,42], they reproduce epithelial branching and acinar lineage development but not yet the full ductal architecture seen in vivo[43,44], and remain essential benchmarks for validating PSC differentiation protocols, despite the ethical and availability constraints that limit their use [45,46]. In practice, the three sources are complementary. In murine studies, adult epithelium is often used, and murine PSC-based approaches include conditional blastocyst complementation[38] and transcription-factor-guided organoid induction[47]. In human studies, cell sources are selected according to the study aim, with patient tissue or iPSCs used for disease modeling and embryonic tissue for development. Drawing on all three, each matched to the question it answers best, gives the most complete platform (Table 1).

Signaling Pathways and Niche Factors

Salivary gland organoids depend on faithfully reconstructing the cellular niche through a balanced set of signals, whose interplay sets the conditions for formation, growth and differentiation[9,13] (Fig. 2).

WNT is the cornerstone of stem cell maintenance and proliferative expansion[14]. Canonical activation, through WNT3A-conditioned medium or the GSK3β inhibitor CHIR99021, stabilizes β-catenin and induces the transcription of genes involved in cell-cycle progression and stem cell identity[14,48], but the level must be titrated: excessive activation impairs differentiation and induces metaplasia, whereas insufficient signaling causes organoid degeneration[13,49]. R-spondins act through LGR receptors to amplify this signal[50]. In the gland, LGR expression is cell-type-specific, with LGR4 predominant in ductal cells and LGR6 marking myoepithelial populations[46,51], and lineage tracing has placed LGR5^{+} cells in a

progenitor pool shared between salivary gland and tongue epithelium[52]. R-spondin 1 is accordingly a standard medium component, synergizing with WNT activators to sustain robust proliferation while preserving stem cell character[14,53,54].

The epidermal growth factor (EGF) superfamily supplies mitogenic and survival signals through EGF receptor (EGFR) and related receptors[55]. EGF is the most common ligand and supports proliferation, maturation and branching, though its concentration and identity bias organoid morphology and lineage output[13]. The importance of signal intensity is well illustrated in glandular systems: in human breastoid cultures, high-affinity EGF drives sustained ERK1/2–RSK activation and an aberrant expansion of the myoepithelial lineage, whereas the lower-affinity ligand amphiregulin, or a reduced EGF dose, restores normal ductal architecture[56]. Neuregulin-1 (NRG1), which signals preferentially through ErbB3/ErbB2 heterodimers and the PI3K–AKT axis rather than the EGFR–MAPK pathway, has therefore been adopted to replace EGF in long-term human salivary organoid culture while preserving glandular identity[13]. The same NRG1/ErbB logic recurs across glandular organoid systems, supporting long-term expansion in mammary cultures[49] and NRG1/HER3-dependent antiandrogen resistance in prostate organoid models[57].

Fibroblast growth factor (FGF) signaling, principally FGF10, governs branching. FGF and WNT act as counteracting pathways, and their antagonism coordinates stalk elongation, branching morphogenesis and end-bud maturation during gland development[58]. FGF10 acts as a mesenchyme-derived signal through FGFR2b, engaging both ERK1/2 and AKT to regulate branching and cell survival[59,60]; in culture, FGF10 supplementation enhances branching complexity, though careful titration is needed to preserve organoid integrity[61].

Several inhibitory inputs must also be modulated. Noggin suppresses endogenous bone morphogenetic protein (BMP) that would otherwise block organoid formation and drive unwanted differentiation[13,62] and the transforming growth factor-β (TGF-β) inhibitor A83-01

suppresses TGF-β–induced epithelial–mesenchymal transition in experimental cell systems[63], and is used in salivary organoid culture13,89. However, TGF-β signaling has context-dependent roles in salivary gland development and tissue remodeling[64], and its dysregulation has been implicated in glandular fibrosis and dysfunction[65].

Notch is where niche control meets the field's central problem. Notch signaling supports stem/progenitor maintenance and regenerative migration[66,67]80, whereas differentiation media containing the γ-secretase inhibitor DAPT can promote expression of acinar markers such as AQP5 and MIST1[13]. In human salivary organoids, DAPT was combined with withdrawal of prostaglandin E2 (PGE2), nicotinamide and CHIR99021; therefore, the differentiation response cannot be attributed to DAPT alone. Although differentiation competence was retained during extended culture, the persistence of acinar identity after withdrawal of differentiation cues and the effects of prolonged Notch suppression on ductal, basal and myoepithelial populations remain insufficiently characterized. Moreover, DAPT inhibits γ-secretase, which processes substrates beyond Notch, so additional substrate-dependent effects cannot be excluded [NEW: Ran et al., 2017]. Hippo signaling, through YAP and TAZ, integrates mechanical cues from the matrix to influence organoid size, shape and branching[68,69].

Culture Systems and Technical Considerations

Culture methods have advanced from simple suspensions to defined systems, and the physical and chemical environment shapes development at every step[9,21]. Extracellular matrix (ECM) choice is the most consequential decision. Matrigel, a laminin- and collagen-IV-rich basement-membrane extract, remains the standard and supplies the cues for three-dimensional organization and branching[61,70], but its undefined composition, batch variability and xenogeneic origin obstruct standardization and clinical translation[71]. The demonstration that a chemically defined, Matrigel-free system yields functional human organoids with improved epithelial

polarity and acinar maturation is therefore significant on two counts: improved translational compatibility and enhanced acinar competence[21]. Defined alternatives are expanding: poly(ethylene glycol) (PEG) hydrogels functionalized with adhesion peptides[72,73], decellularized salivary ECM that retains tissue-specific cues[74], and alginate–gelatin composites that support acinar expansion at low cost with easy cell recovery[75]. Experimental studies further illustrate the importance of matrix properties: enzymatically degradable PEG hydrogels promote acinar-associated characteristics in primary salivary gland cells[73], while reversible thermo-ionically crosslinked hydrogels support the expansion of functional human salivary acinar spheroids[75]. Together, these findings suggest that matrix composition, degradability and mechanical context influence epithelial phenotypes, although the independent contributions of these factors remain incompletely resolved.

Medium composition balances growth against differentiation. Adult-tissue formulations combine EGF, R-spondin 1 and a WNT activator with FGF10, Noggin, a TGF-β inhibitor and the ROCK inhibitor Y-27632[13,14,62,76]. Beyond composition, the timing of factor addition and withdrawal is decisive, particularly for PSC-derived organoids that require sequential pathway activation to specify lineage and mature[47].

Passaging and storage introduce additional sources of variability. Organoids are typically passaged mechanically or enzymatically at protocol-dependent intervals[11,13,77], and dissociation can disrupt architecture and alter the representation of epithelial subtypes[23]. Cryopreserved cultures therefore require post-thaw assessment of viability and epithelial composition[23]. Because acinar populations can be clonally heterogeneous in another exocrine organ, the pancreas[78], longitudinal monitoring of cellular composition across the culture lifecycle is essential.

As a two-dimensional alternative, chemical reprogramming culture uses a feeder- and Matrigel-free defined medium with small-molecule inhibitors to expand human salivary basal

progenitors while preserving ductal progenitor identity and multilineage potential[79]. By supplying defined progenitors under simple, xeno-reduced conditions, it offers a cell source that feeds directly into cell-based therapy, a strategy we return to in the section on regeneration.

CHARACTERIZATION AND FUNCTIONAL VALIDATION

Structural Analysis and Molecular Markers

Validating salivary gland organoids begins with their structural organization and molecular composition[14,23,77,80]. Characterization draws on immunofluorescence, histological staining, bulk and single-cell expression profiling, ultrastructural analysis, and functional readouts such as mucin/glycoprotein staining, amylase activity and calcium mobilization[14,61,77].

The native epithelium is compartmentalized into luminal ductal and acinar cells and abluminal basal and myoepithelial cells, so well-differentiated organoids should be assessed for lumen formation, polarized ductal markers, acinar/secretory induction and abluminal features[14,61,77]. Acinar cells are marked by AQP5, BHLHA15 and AMY, ductal cells by KRT7, KRT8/18 and KRT19, basal cells by KRT5, KRT14 and TP63, and myoepithelial cells by co-expression of KRT14 and ACTA2[14,36,61,77,79]. AQP5 localization is especially informative: apical enrichment in acinar-like cells signals the epithelial polarization required for vectorial water movement and fluid secretion[81,82].

A range of markers, including ITGB1/CD29, CD44, CD24, CD49f, EpCAM, SOX9, SOX10, KRT5/KRT14, TP63 and context-dependent KIT/CD117, has been linked to stem/progenitor-like states, but these are best read as context-dependent indicators of regenerative potential and lineage bias rather than markers of any universal salivary gland stem cell[23,41,80,83-85]. Their spatial localization is nonetheless informative: in intestinal and gastric organoids, marker mapping has resolved stem cell niches and guided reverse identification of native progenitor populations[29,86,87], an approach that may help settle debate over salivary

stem/progenitor identity[23,80,88]. The transcription factors SOX2, SOX9, and SOX10 mark developmental, progenitor, or secretory-lineage programs relevant to self-renewal and differentiation capacity in salivary gland models[41,42,85]. Single-cell transcriptomics reinforces this, showing model- and stage-dependent mixtures of ductal, basal, progenitor-like, pro-acinar and myoepithelial-like cells rather than a single hierarchy[36,77,80]. Histology and ultrastructure add the spatial view, with periodic acid–Schiff (PAS) and Alcian blue staining marking mucin-producing cells and electron microscopy resolving serous and mucous acinar-like cells with electron-dense or electron-pale vesicles alongside myoepithelial cells[14,77].

Functional Assays

Functional validation extends beyond structure to the processes that define glandular function: secretory capacity, ion transport and neurotransmitter responsiveness[14,89] (Fig. 3).

Calcium assays probe the signaling underlying secretion using fluorescent indicators such as Fluo-4 to track cytoplasmic Ca^{2+} after secretagogue stimulation[89,90]. In salivary organoids, carbachol and adenosine triphosphate (ATP) elicit calcium transients that indicate intact secretory signaling[13].

Swelling assays offer a visually intuitive readout of fluid secretion. Differentiated salivary organoids can swell in response to carbachol, isoproterenol and vasoactive intestinal peptide[13], while carbachol- and forskolin-induced swelling has also been demonstrated in human salivary gland-derived organoids[89]. Forskolin-induced swelling was first established in intestinal organoids as a cystic fibrosis transmembrane conductance regulator (CFTR)-dependent measure of epithelial anion and fluid transport[91]. The magnitude and kinetics of swelling, tracked by time-lapse imaging, report integrated ion-channel, transporter and water-channel activity.

Amylase production is a defining acinar readout because the enzyme is a major

salivary component[14]. It can be assessed by expression, intracellular localization or enzyme-activity measurements and interpreted alongside other secretory markers[13,14,90]. Chloride-transport function can be assessed using complementary pathway-specific readouts. CFTR-dependent forskolin swelling [91] and ANO1/TMEM16A-dependent salivary secretion[92] provide relevant pathway benchmarks. Halide-sensitive fluorescent indicators, when combined with appropriate stimulation or inhibition, may provide a complementary readout of chloride flux, although direct validation of such assays in salivary organoid systems remains limited. Ussing-chamber recordings on organoid-derived monolayers can provide more detailed electrophysiological measurements when technically feasible.

Species Differences: Murine versus Human Organoid Models

This divergence is the developmental tension in concrete form: the degree of acinar maturation a system achieves sets the ceiling on the secretory, pharmacological and regenerative functions it can later support. The gap between murine and human systems is the field's defining methodological challenge, with direct consequences for translational research[13,93-95] (Fig. 4). It reflects not only species-specific gland anatomy and secretory specialization but also divergent responses to niche factors and extracellular matrix, together with the effects of donor age, pathology, and culture-induced selection.

Murine salivary gland organoids can exhibit acinar, ductal and myoepithelial lineage features under optimized differentiation conditions, although the extent of differentiation varies with gland of origin and culture protocol[13,14,61]. They form lumenized or branched structures expressing gland-associated secretory markers such as AQP5, BHLHA15, AMY and mucin-related proteins, depending on the gland of origin and protocol[13,14,61]. Differentiation outcomes may also vary with passage history and matrix conditions, with batch-variable matrices introducing an additional source of experimental variability[71]. Ultrastructural analyses confirm secretory maturation, revealing differentiated serous and mucous acinar-like cells with secretory vesicles and associated myoepithelial cells[14], while functional assays demonstrate

secretagogue-responsive activity, including neurotransmitter-induced swelling, amylase secretion and calcium mobilization[13,14,96].

Human organoids, by contrast, are typically ductal- or basal/progenitor-enriched under standard expansion, forming cystic, solid, dense or branched structures with basal cells expressing KRT5 and KRT14 and luminal cells expressing KRT7 and KRT19; robust expression of acinar markers such as AQP5, BHLHA15, AMY1A and lactoferrin remains protocol-dependent[13,77,79,89]. Although WNT/EGF-rich expansion conditions may favor proliferative ductal and basal states, their contribution to limited acinar maturation relative to starting cell composition, gland of origin, passage-dependent selection, ECM composition and other niche cues remains incompletely resolved[13,14,77.13,21,23,77]

Several strategies have been used to address this limitation. Targeted differentiation approaches, including Notch inhibition with DAPT, reduction of expansion-associated supplements and modulation of ALK/TGF-β signaling, have promoted acinar or secretory marker expression and functional responses[13,77,89], while matrix-based approaches such as collagen-based organoids, hyaluronate hydrogels and salivary gland-derived decellularized ECM aim to improve epithelial organization and clinical applicability[77,97,98]. Chemically defined, xeno-free or Matrigel-free systems have gone furthest, achieving multilineage differentiation with preserved epithelial heterogeneity and a single-cell basal-to-ductal-to-acinar trajectory[21], yet complete acinar maturation and full recapitulation of native secretory output remain unresolved[21,77].

These differences are likely multifactorial, but the relative contributions of intrinsic species biology and culture-dependent factors remain unresolved. Potential contributors include species-specific gland architecture and transcriptomic programs, stem/progenitor states, divergent WNT, EGF, FGF, Notch and TGF-β/ALK requirements, matrix-dependent responses, and donor variables such as age, disease, inflammation and prior injury[13,23,93-95,99].

Many human cultures originate from adult, donor-variable, ductal/basal-enriched populations and may require additional niche cues for acinar maturation; however, determining whether these factors account for the observed species differences will require comparisons under more closely matched culture and differentiation conditions. The widespread use of mouse-sarcoma-derived Matrigel further introduces undefined, batch-variable ECM, although direct evidence that murine matrix alone impairs human acinar maturation remains limited[77,98].

These properties define complementary roles. Murine organoids, experimentally tractable and readily differentiated, are well suited to studying gland morphogenesis, lineage specification, secretagogue responses and injury–repair mechanisms[13,14,61]; human organoids, despite their ductal/progenitor bias, are indispensable for translation because they preserve patient-specific genetic backgrounds for modeling disease susceptibility, radiation injury, tumor biology and personalized therapeutic responses[13,23,77]. The two systems are most powerful when used together (Table 2). For secretory pharmacology and regenerative therapy, the maturity of the epithelial compartment also determines how closely experimental readouts approximate native gland function.

DISEASE MODELING I: TUMOR ORGANOIDS AND PRECISION ONCOLOGY

The Unique Landscape of Salivary Gland Tumor Diversity

Salivary gland cancers comprise 21 malignant epithelial tumor types in the fifth edition of the WHO Classification of Head and Neck Tumours, each with distinct histopathology, molecular profile, biological behavior and clinical outcome[7,100]. The classification increasingly incorporates recurrent molecular alterations, particularly tumor-type-specific gene fusions, into diagnostic definitions and, in selected subtypes, therapeutic decisions[7]. This heterogeneity translates into wide variation in disease course and treatment response, which frustrates broadly applicable strategies and makes the case for subtype-specific precision oncology. Because most

subtypes are rare, biomarker studies and subtype-specific clinical trials are frequently underpowered[17,101-103]. Modeling this diversity brings the architectural tension to the fore: faithful patient-derived organoids must preserve subtype identity while resolving neoplastic epithelium from the stromal and immune compartments that current cultures largely omit.

The spectrum runs from the more common mucoepidermoid carcinoma (MEC) and adenoid cystic carcinoma (ACC) to rare variants such as epithelial–myoepithelial carcinoma (EMC), carcinoma ex pleomorphic adenoma (CaExPA) and polymorphous adenocarcinoma[7,100]. Each carries characteristic morphology and molecular alterations that reflect distinct cellular origins and pathogenetic mechanisms. MEC frequently harbors CRTC1/3–MAML2 rearrangements that activate CREB-associated transcriptional programs and AREG–EGFR signaling[104-106]; ACC shows cribriform, tubular and solid growth patterns with MYB–NFIB or MYBL1–NFIB fusions[107-109], alongside recurrent alterations in chromatin-remodeling complexes, DNA-damage-response pathways and NOTCH signaling [110-112] (Fig. 5 and Table 3).

Patient-Derived Organoids: Establishment and Characterization

Patient-derived organoids (PDOs) model individual salivary gland tumor subtypes while preserving the key histopathological, molecular and differentiation features of the parental tumor, and collections spanning multiple subtypes are beginning to provide experimentally tractable models of a disease spectrum that has been poorly represented in conventional cell culture systems[16,17,101,113].

Establishment typically begins with fresh surgical specimens, followed by controlled mechanical and enzymatic dissociation designed to preserve viable tumor epithelial clusters while minimizing single-cell stress and stromal contamination[114,115]. Reported success rates vary substantially and depend on how establishment is defined, including short-term organoid

formation, serial passaging capacity and recovery after cryopreservation. In one systematic effort, viable short-term cultures were generated from 7 of 37 salivary gland cancer specimens[16]. Such selective establishment may favor tumors that grow readily in culture while underrepresenting other subtypes or cell populations, potentially limiting the generalizability of PDO-based findings. Contributing factors include specimen size and cellularity, necrotic burden, ischemic and processing time, intrinsic growth kinetics, prior treatment exposure and the proportion of viable tumor epithelium[116,117].

A particularly important challenge is the admixture or overgrowth of residual normal salivary epithelial, basal/myoepithelial or stromal populations during early passage[118,119], an issue that is especially complex in biphasic tumors, where neoplastic luminal and abluminal populations must be distinguished from residual normal ductal, basal and myoepithelial cells[120,121]. Rigorous characterization should therefore combine histomorphological comparison with the parental tumor, immunophenotypic assessment of lineage-specific markers, and molecular authentication using fusion testing, targeted sequencing, copy-number profiling, whole-exome sequencing or RNA sequencing[16,101,113]**(Box 1->Table 4)**. Because serial passaging can drive clonal selection, loss of minor cell populations and transcriptomic drift, particularly in slow-growing tumors such as ACC, passage number should be documented and key histological and molecular features reassessed at defined intervals, with retention of defining alterations such as MYB–NFIB regarded as a key criterion of disease relevance[16,101]. Although long-term establishment remains inconsistent, PDOs have been reported from multiple subtypes, including ACC, MEC, salivary duct carcinoma (SDC), myoepithelial carcinoma and secretory carcinoma, demonstrating feasibility while underscoring the need for subtype-specific culture conditions and standardized authentication criteria[16,101,102,113].

Precision Medicine: Drug Screening and Therapeutic Stratification

For localized disease, surgery with risk-adapted radiotherapy remains the treatment backbone; for recurrent or metastatic disease, however, systemic options remain limited and increasingly depend on molecular profiling and subtype-specific targets. In this context, PDO-based drug screening adds a functional validation layer to genomic precision oncology by testing whether molecular alterations translate into actual drug sensitivity in patient-derived tumor cells[113,122,123]. This is particularly valuable for salivary gland cancers, where treatment options are limited and individual subtypes are too rare for conventionally powered trials[124,125].

Organoid-based platforms evaluate multiple compounds and dose ranges in parallel, with screening capacity determined by organoid expansion and assay format, and ATP-based luminescence assays or metabolic dyes quantify organoid viability after treatment[123,126]. Drug concentrations and exposure schedules should be selected with reference to clinically achievable pharmacokinetics rather than arbitrary in vitro dose ranges[126,127]. Building on genomic diagnosis, organoid-based functional testing can confirm, refine or challenge predicted therapeutic vulnerabilities, as demonstrated by tropomyosin receptor kinase (TRK) inhibitor testing in ETV6–NTRK3 fusion-positive secretory carcinoma[102] and by genotype-oriented pathway testing in salivary gland cancer organoids[113]. For human epidermal growth factor receptor 2 (HER2)-amplified or androgen receptor (AR)-positive SDC, PDOs may support patient-specific testing of HER2-directed agents, androgen receptor inhibition and selected combination regimens. This functional assessment helps determine whether a molecularly or immunophenotypically defined target represents a true therapeutic dependency rather than merely a detectable biomarker[113,125,128]. The applications of PDOs also differ by tumor subtype. In SDC, organoids have supported genotype-informed testing of AR antagonism and PI3K–mTOR/RAS–MAPK pathway inhibitors, although incomplete retention of HER2 amplification in some cultures highlights the need for molecular reauthentication[113]. In CRTC1/3–MAML2-positive MEC, AREG–EGFR signaling provides a candidate target for

PDO-based screening, based on findings from cell-line and xenograft studies rather than validated MEC PDO drug responses[104–106]. ACC organoids may facilitate studies of MYB/MYBL1 fusion-associated biology and molecularly selected NOTCH-pathway vulnerabilities, although slow growth and culture-associated selection can complicate longitudinal studies and drug screening[101,107–113]. For rarer subtypes, establishing representative PDO models and verifying tumor identity using subtype-appropriate authentication criteria (Table 4) remain prerequisites for meaningful therapeutic evaluation[101,113].

Beyond single agents, organoids provide a preclinical platform for prioritizing rational drug combinations before clinical evaluation, an especially useful function in rare cancers for which large combination trials are difficult to conduct[113,123]. Although salivary gland cancer-specific radiosensitivity data remain limited, organoids from other head and neck cancers provide proof of principle that patient-derived models can reproduce interpatient variation in radiation response[129,130]. Across sufficiently large PDO cohorts, integrating drug-response profiles with genomic and transcriptomic data may identify predictive biomarkers and define molecularly informed responder subgroups, which can then be evaluated in registries, basket trials or prospective clinical studies[17,113].

Clinical implementation still faces practical challenges, including establishment timelines compatible with treatment decision-making, standardized culture and assay protocols, tumor-cell authentication, sufficient assay success rates and prospective clinical validation, together with the selection bias introduced when particular subtypes or rapidly growing clones establish preferentially[123,127,131]. When establishment and drug testing take weeks to months, PDO testing may be more feasible for informing subsequent treatment lines than urgent initial decisions in rapidly progressing disease. Prospective studies should report the proportion of patients receiving interpretable results within the relevant treatment decision window[123,127,131]. A registered prospective, multicenter observational study (NCT06686979;

planned enrollment, n = 40) is evaluating concordance between PDO drug sensitivity and clinical treatment outcomes in recurrent or metastatic salivary gland cancer. Patients receive guideline- or physician-directed systemic therapy independently of PDO testing, while biopsy-derived organoids undergo parallel drug-sensitivity assessment. The registered primary outcomes are objective response rate and disease control rate over two years, with secondary outcomes including PDO establishment and testing success, ex vivo tumor growth inhibition, progression-free survival and overall survival. Although designed as a multicenter study, the public registry currently identifies Huashan Hospital as the only named study institution, leaving the number of distinct participating centers unclear [N1]. With prospective validation and protocol standardization, PDO-based functional testing may complement genomic, pathological and clinical information to support more rational therapeutic selection for patients with limited treatment options. Accordingly, translational relevance depends on both preserving disease-relevant tumor features and obtaining interpretable results within the treatment decision window, while incorporating stromal and immune influences remains a further architectural challenge.

DISEASE MODELING II: SJÖGREN'S DISEASE AND AUTOIMMUNE ORGANOIDS

Sjögren's disease (SjD) is a prototypical autoimmune condition in which progressive dysfunction and destruction of the salivary and lacrimal glands cause debilitating sicca symptoms and, in a subset of patients, systemic manifestations[132-134]. It is the clearest expression of the architectural tension: because the disease is driven by immune–epithelial and stromal–epithelial crosstalk, epithelial-only organoids capture only part of its biology, and reconstituting these interactions could extend disease modeling beyond the epithelial-intrinsic processes captured by current organoids[20,135,136]

Epithelial-Intrinsic Dysfunction in Sjögren's Disease

Organoids derived from SjD patient biopsies reproduce disease-relevant epithelial features even without immune cells, including persistent interferon and Janus kinase–signal transducer and activator of transcription (JAK–STAT) signaling, altered epithelial phenotypes and impaired functional responses[20,137]. Long-term organoids from healthy and SjD human minor (labial) glands retain native gland-specific phenotype and secretory function, and single-cell transcriptomics resolves distinct phenotypic differences between healthy- and patient-derived organoids, indicating that disease-associated epithelial programs persist in culture[137]. Tight-junction disruption, a recognized abnormality of SjD epithelium linked to inflammatory cytokine exposure, has yet to be directly demonstrated in patient-derived organoids[132,138]. Because these intrinsic alterations are quantifiable in vitro, epithelial-only organoids provide a controlled system to quantify disease-associated epithelial phenotypes and therapeutic responses and to define the epithelium's autonomous contribution to pathogenesis, independent of ongoing immune attack[20,136,137].

Chimeric Organoid Models: Stromal-Epithelial and Immune-Epithelial Crosstalk

Direct studies of human SjD-derived salivary gland organoids have primarily characterized epithelial-intrinsic dysfunction[20,137], whereas immune and stromal cell populations have been characterized in native SjD gland tissue[139]. Multimodal profiling of human SjD glands specifies which partners to reconstitute: single-cell and spatial analyses identify cytotoxic $GZMB^{+}GNLY^{+}$ $CD8^{+}$ T cells as the dominant expanded effector across autoantibody subgroups and nominate complement- and chemokine-expressing $THY1^{+}$ fibroblasts as stromal hubs of inflammation[139], thereby defining the immune and stromal components for causal testing in chimeric organoids.

Because epithelial-only cultures cannot capture the interactions that sustain autoimmune injury, recent work has moved toward immune-integrated systems, with proof of principle emerging largely from other tissues. In the intestine, air–liquid-interface organoids grown directly from endoscopic biopsies preserve epithelium, native mesenchyme and tissue-resident immune cells as a single unit and reproduce antigen-driven, immune-mediated epithelial destruction, establishing a general framework for organoid modeling of autoimmunity[140]. A complementary strategy integrates autologous tissue-resident memory T cells into epithelial organoids, where they engraft and, upon activation, drive cytotoxic injury that can be pharmacologically modulated[141]. Applied to the salivary gland, analogous co-culture and chimeric strategies that combine organoids with selected immune populations, inflammatory cytokines or patient-derived immune components could model autoantigen presentation, interferon responses and functional decline that are difficult to isolate in vivo[132,135].

Inflammatory epithelial–stromal interactions have been implicated in salivary gland dysfunction in SjD, providing a rationale for investigating whether pathways such as tumor necrosis factor-α (TNF-α)/nuclear factor-κB (NF-κB) influence acinar differentiation[132,135]. Recent work using murine chimeric salivary gland organoids provides direct experimental evidence that multicellular stromal environments can be incorporated into salivary organoid models. In this system, embryonic submandibular stromal cells were combined with postnatal epithelial organoids, introducing fibroblast, macrophage/monocyte-like, endothelial and neuronal populations [NEW: Yoon et al., JDR, in press]. Single-cell analyses and stromal fractionation identified PTPRC-enriched stromal cells as a candidate source of TNF-superfamily ligand transcripts, while TNF-α exposure reduced selected acinar-associated markers and dexamethasone or NF-κB pathway inhibition increased their expression under the tested conditions [NEW: Yoon et al., JDR, in press]. These findings provide a tractable model

for investigating inflammation-associated stromal–epithelial interactions but do not establish direct cell-specific causality or fully restored acinar maturation. Further validation in patient-derived human SjD organoids is needed to determine whether comparable inflammatory interactions contribute to impaired regeneration and loss of secretory identity.

Precision Immunology: Patient-Specific Therapeutic Modeling and Biomarker Discovery

Next-generation immuno-organoid and assembloid systems aim to reconstruct autoimmune injury with greater cellular and spatial complexity by incorporating autologous, tissue-resident immune compartments while preserving endogenous immune diversity[142]. If developed for salivary gland disease, assembloids combining epithelial organoids with autologous immune cells and stromal fibroblasts could enable dynamic study of immune–epithelial crosstalk and, crucially, patient-specific testing of immunomodulatory drugs[135,143]. No mature human salivary gland immune–epithelial assembloid has yet been reported; proposed immune-integrated models for SjD therefore draw on approaches developed in other tissues[140,141]. The ability of patient-derived SjD organoids to report pharmacological responses has already been demonstrated using the JAK inhibitor tofacitinib, which enhanced their proliferation by suppressing JAK–STAT signaling[137], providing a concrete example of organoid-based therapeutic evaluation. Such models could also help dissect disease mechanisms and support therapeutic target nomination and validation[141].

These considerations matter given the clinical heterogeneity of SjD. Patient glands stratified by autoantibody status carry distinct signatures characterized by an activated interferon program in anti-SSA-positive disease and enhanced TGF-β signaling in anti-centromere-positive disease, implying different actionable pathways among serologic subgroups[139]. Reconstructing these subgroup-specific signatures in patient-derived organoids and linking clinical clustering to functional readouts could support patient-specific prediction

of therapeutic responses and biomarker discovery, thereby extending the precision oncology paradigm into a precision immunology framework for autoimmune salivary gland disease. The architectural challenge is therefore to reconstruct disease-relevant immune–stromal interactions without losing the experimental resolution afforded by epithelial-only organoids.

DISEASE MODELING III: RADIATION-INDUCED SALIVARY DYSFUNCTION

Radiation-induced salivary dysfunction is a frequent and debilitating consequence of head and neck cancer treatment. Most patients receiving radiotherapy develop some xerostomia, though estimates of clinically significant hyposalivation vary with dose, irradiated gland volume, treatment modality, salivary endpoint and follow-up, and the resulting chronic oral complications severely affect quality of life[144,145]. Organoid models and salivary tissue-chip platforms capture selected acute and chronic features of radiation injury, supporting mechanistic study, radioprotective screening and regenerative therapy testing[77,146-148].

Acute Radiation Injury: DNA Damage, Apoptosis, and Secretory Dysfunction

During the acute phase, within the first week after irradiation, mouse and human salivary organoid-derived cells have been used to assess DNA-damage responses and organoid-forming survival[148], whereas murine salivary tissue chips have reproduced dose-dependent γH2AX and 53BP1 responses benchmarked against irradiated mouse gland tissue[147]. Other mechanisms implicated in salivary radiation injury include reactive oxygen species (ROS) accumulation, mitochondrial dysfunction**,** aberrant calcium signaling**,** p53-dependent apoptosis, epithelial tight-junction disruption and altered localization or expression of the water channel AQP5[144]; these features have not all been demonstrated in each model. Double-strand-break markers such as γ-H2AX, together with calcium and secretory assays, provide complementary rather than interchangeable measures of early injury[147,148]**.**

Chronic Sequelae: Cellular Senescence and Niche Impairment

Chronic-phase studies in mouse and human salivary glands and murine organoids capture long-term sequelae: cellular senescence marked by p16INK4a upregulation, persistent DNA-damage signaling and a senescence-associated secretory phenotype (SASP), alongside progressive impairment of the stem/progenitor niche[146,149]. SASP factors, including pro-inflammatory cytokines (IL-6, IL-8), matrix metalloproteinases and extracellular vesicles, may act non-cell-autonomously, propagating senescence through paracrine signaling and TGF-β-associated stromal fibrosis; epithelial-only organoids cannot reproduce the full stromal response. Although many experimental studies use a single acute high dose, fractionation can produce endpoint-dependent effects: four daily fractions of 0.25 Gy reduced organoid-forming survival more than a single 1 Gy dose in mouse salivary organoid-derived cells[148], whereas single doses of 10 or 15 Gy and three fractions of 5 Gy produced broadly comparable acute injury at day 3 in an in vivo mouse study [NEW: Johnson et al., 2024]. These results do not establish equivalence in long-term gland function or clinical radiotherapy. Spatial dose distribution is also important, because sparing stem-cell-rich gland regions preserves salivary output[150]. However, conventional organoids do not recapitulate intact-gland anatomy or the spatial dose gradients needed to assess region-specific sparing.

Organoid Platforms for Radioprotective and Senolytic Drug Screening

Murine salivary tissue chips have been used to assess WR-1065 radioprotection, with DNA-damage responses benchmarked against irradiated mouse gland tissue[147]. A subsequent high-content tissue-chip screen evaluated 438 approved compounds, identified 25 double hits and tested three candidates in irradiated mice [NEW: Piraino et al., 2025]. Separately, senescence-targeted treatment improved organoid-forming efficiency in irradiated murine salivary

organoids, while ABT263 preserved salivary function in irradiated mice[146]. These studies demonstrate screening feasibility but highlight the need to establish assay robustness, throughput, reproducibility across preparations and batches, and dose-response performance against appropriate controls. DNA-damage and senescence-marker responses should be assessed alongside functional endpoints, including organoid formation and salivary flow; reduced p16, p21 or SASP-marker expression alone does not demonstrate selective senescent-cell clearance. Because conventional epithelial organoids lack stromal and immune compartments, multicellular co-culture or tissue-chip systems are needed to model this complexity and its pharmacological modulation. In irradiated mouse recipients, human salivary organoid xenografts demonstrated engraftment, whereas murine organoid allografts increased salivary secretion[77]. These preclinical findings support regenerative potential, although engraftment durability, long-term safety and scalability remain unresolved. Coupled with salivary-specific assays of amylase secretion, calcium transients and tight-junction integrity, these platforms bridge phenotypic screening and functional pharmacology.

FUNCTIONAL PHARMACOLOGY PLATFORMS

Building on their use in radioprotective and senolytic screening, salivary gland organoids are emerging as functional pharmacology platforms for agents acting on epithelial secretion, tissue regeneration and neurosecretory signaling. This shifts the organoid from a disease model to a quantitative assay of agonist responsiveness, secretory competence, drug potency and pharmacodynamic effect, although regulatory or industrial use will require assay qualification, benchmarking against accepted reference methods and cross-laboratory reproducibility.

Translational Pharmacology for Xerostomia Therapeutics

One potential application is the preclinical evaluation of agents that modulate salivary

neurosecretory function. Cholinergic agonists such as pilocarpine and cevimeline, established treatments for xerostomia and salivary hypofunction, may be assessed using organoid secretory readouts to examine receptor responsiveness, residual secretory capacity and concentration-dependent effects[13,145,151]. Pilocarpine-induced swelling has been demonstrated in organoids derived from human labial salivary gland biopsies[20], whereas comparable organoid-based evaluation of cevimeline remains to be established. These assays provide functional pharmacodynamic readouts rather than clinical dose optimization, since systemic tolerability, autonomic side effects, pharmacokinetics and residual gland volume still govern patient response.

Novel secretagogues in development, including muscarinic-targeted agents, purinergic modulators, ion-channel regulators and compounds restoring calcium-dependent secretion, need platforms that recapitulate integrated epithelial secretion[13,151,152]. Because salivary organoids report calcium signaling, swelling and gland-type-specific secretory markers, they serve as a practical intermediate between reductionist receptor assays and in vivo salivation studies[13,89]. This intermediate role is clinically motivated: cholinergic secretagogues act only where residual functional acinar tissue remains and so offer limited benefit after extensive radiation- or autoimmune-mediated acinar loss. Organoid platforms that couple secretagogue testing with regenerative readouts could therefore help identify agents or agent–cell combinations that restore secretion in glands with depleted acinar reserves, a population poorly served by current pharmacotherapy.

Botulinum Neurotoxin Potency Testing

A prominent neuroglandular application is botulinum neurotoxin (BoNT) potency testing, which traditionally relies on the mouse median lethal dose (LD50) assay, an approach that raises ethical concerns, consumes large animal numbers and is subject to variability in injection, endpoint and

interlaboratory implementation that complicates standardization[153-155]. Validated cell-based potency assays (CBPAs), particularly those measuring BoNT uptake and cleavage of synaptosomal-associated protein 25 (SNAP25) in neuronal systems, have achieved regulatory acceptance for selected BoNT/A products[156-159]. But these model neuronal intoxication rather than gland-specific neuroepithelial responses, so salivary organoids are best positioned as complementary gland-specific pharmacodynamic platforms rather than replacements.

This application has direct clinical grounding. Beyond its neuromuscular indications, botulinum neurotoxin is injected into the parotid and submandibular glands to treat sialorrhea in Parkinson's disease, amyotrophic lateral sclerosis, cerebral palsy and other neurological conditions, where the therapeutic aim is to reduce glandular secretory output rather than to block skeletal muscle[160,161]. Mechanistically, BoNT inhibits acetylcholine release at cholinergic terminals of the neuroglandular junction, so an assay that reads the glandular secretory response measures the pharmacodynamic endpoint most relevant to this indication. In murine salivary gland organoid (SGO)–Neuro-2a (N2a) neuronal co-cultures, neuronal processes contact the organoid surface, acetylcholine or neostigmine stimulation increases the expression of secretory markers such as *Aqp5* and *Bhlha15*, and BoNT/A treatment reverses these responses, supporting a neuroepithelial organoid-based BoNT assay[162]. Because neuronal cell-based potency assays were developed and optimized for neuromuscular products, they do not directly capture this gland-specific pharmacology; the next step is therefore to benchmark these readouts against established potency standards, test whether they add predictive value, and demonstrate reproducibility across laboratories and BoNT serotypes.

Organoid Platform for Fibrosis and Inflammation

Comprehensive pharmacology must also address the fibrotic and inflammatory microenvironments driving chronic dysfunction in radiation injury and Sjögren’s disease.

Radiation-induced senescence and cytokine signaling can impair the regenerative niche[149], providing a rationale for testing cell-free therapeutics, particularly extracellular vesicles (EVs). Evidence from intestinal[163], hepatic[164] and retinal[165] models supports the broader anti-inflammatory and antifibrotic potential of EVs but does not constitute direct salivary gland evidence. In the salivary gland, CD9-enriched EVs from chemically reprogrammed basal progenitors attenuated fibrosis in human salivary organoid assays and a mouse duct-obstruct model[166]. In organoid-based assays, EVs could be evaluated for their ability to reduce inflammatory or fibrotic signaling and restore epithelial progenitor function. **However, assessment of stromal fibrosis or immune modulation requires the relevant non-epithelial compartments.** A remaining limitation is scalability: multicellular organoids carry inherent complexity and batch variability, and nonstandardized ECM and inconsistent immune-cell ratios constrain high-throughput use[167,168].

Regulatory Implications and the 3R Framework

These advances answer the growing demand for animal-free or animal-reducing methods and align with the Replacement, Reduction and Refinement (3R) principles increasingly emphasized by regulators and pharmacopeial bodies[153]. The European Pharmacopoeia already permits BoNT potency testing by LD50 or by methods validated against it, and product-specific CBPAs show that non-animal testing can meet regulatory expectations for selected BoNT products[156,158,159]. Salivary organoid–neuronal co-culture assays may provide complementary readouts of gland-specific pharmacodynamics, but, unlike validated product-specific neuronal CBPAs, they remain exploratory and are not accepted as regulatory lot-release potency assays for BoNT. Their further development requires standardized co-culture conditions, benchmarking against established potency methods and demonstrated cross-laboratory reproducibility; organoid-based evaluation of complex biologics such as EVs also remains investigational.

REGENERATIVE THERAPEUTICS

Organoid Transplantation: From Preclinical Validation to Clinical Trials

Regeneration is where the three tensions converge on the clinic: a transplantable product must achieve acinar maturation (developmental), integrate with stromal, vascular and neural compartments (architectural), and be manufactured reproducibly under regulatory constraints (translational). Transplantation illustrates this convergence most directly, and has progressed from proof-of-concept studies to early clinical evaluation, representing one of the most advanced directions in the field[15,21,22,169]. Preclinical work from several groups has established the feasibility of organoid or stem/progenitor cell transplantation to restore gland function. Human salivary stem/progenitor cells grown as salispheres or organoids self-renew, differentiate into multilineage organoids, engraft in irradiated mouse glands and partially restore saliva production in xenotransplantation models[15]. Orthotopic transplantation of human organoids into the submandibular glands of non-obese diabetic (NOD) mice, a Sjögren's disease model, significantly improved stimulated salivary flow, increased acinar AQP5 expression and promoted structural integration within the host ductal network[21]. Lymphocytic focus scores were unchanged, suggesting that improvement was driven by epithelial restoration rather than broad immunosuppression[21]; green fluorescent protein (GFP) tracing confirmed survival and ductal integration, and proteomic analysis of saliva detected human-specific salivary proteins, providing direct evidence of secretory activity from engrafted cells[21]. Ectopic transplantation under the renal capsule further showed that human organoids can undergo multilineage differentiation and form gland-like structures with mucous-rich lumens, secretory features and vascular integration even outside the native niche, and co-transplantation with embryonic murine salivary mesenchyme enhanced AQP5- and MUC7-positive acinar-like differentiation, consistent with mesenchyme-associated acinar maturation in this ectopic mouse model[21]. Even so, these

observations do not establish mature neural or immune integration, nor physiological drainage into the oral cavity.

Comparable feasibility holds across matrices and species. In a recent study, human salivary stem cells expanded as Matrigel- or collagen-based human salivary gland organoids (hSGOs) and differentiated into acinar and ductal lineages; collagen-based hSGOs engrafted in mouse recipients and expressed mucins such as MUC5B, while allogeneic transplantation of murine SGOs increased salivation in a radiation-induced xerostomia model[77]. As a foundational resource, a clinically annotated salivary tissue-organoid biorepository from 208 donors has been established at Mayo Clinic, supporting stem/progenitor characterization, regenerative assay development and future translational studies[23] (Table 4).

Clinical translation has advanced most clearly in the RESTART program at the University Medical Center Groningen, where autologous organoid-derived cells are being evaluated to prevent or treat radiation-induced xerostomia after head and neck cancer therapy[22,169]. RESTART (NCT04593589) is an open-label, single-arm phase I trial at one center, with a planned enrollment of 18 adults undergoing surgery for oral cavity squamous cell carcinoma followed by postoperative (chemo)radiotherapy involving the remaining submandibular gland. Patients with systemic diseases affecting salivary function, including Sjögren's disease, are excluded.

Salivary tissue is obtained before radiotherapy, stem/progenitor cells are expanded ex vivo as organoid-derived cell products under good manufacturing practice (GMP)-compatible conditions, and the product is reintroduced after treatment [22]. Public participant information issued in 2020 describes injection into the remaining irradiated submandibular gland under local anesthesia approximately four weeks after radiotherapy [NEW: UMCG participant information]. The registered primary outcomes are adverse events during the first year and salivary flow at 6 and 12 months, with secondary assessments including patient-reported outcomes and oncological follow-up extending to five years [NEW: NCT04593589].

The published manufacturing process includes cryopreservation of an intermediate cell product, subsequent expansion, and release testing for viability, sterility and endotoxins[22]. The specified range of 5×10^4–5×10^6 cells per vial represents a final-product manufacturing specification rather than a confirmed participant-level administered dose. The final formulated product has a reported refrigerated shelf life of four hours, distinct from the storage of cryopreserved intermediate cells[22]. Clinical outcomes from RESTART have not yet been publicly reported.

Together, efforts in Europe, South Korea and North America indicate growing momentum rather than definitive efficacy. Evidence for graft persistence and functional contribution remains limited to relatively short-term preclinical observations. Human salivary organoids transplanted orthotopically into NOD mice were assessed at seven weeks, with evidence of host ductal integration and human-specific salivary proteins[21]. In a separate radiation-injury model, murine organoid allografts improved salivary secretion through eight weeks after transplantation, although donor-cell detection varied among recipients[77]. Human organoid grafts beneath the renal capsule formed gland-like structures over twelve weeks, but this ectopic transplantation site cannot demonstrate drainage into the oral cavity[21]. Together, these findings support regenerative feasibility without establishing durable, patent ductal drainage and sustained secretion in human recipients.

Autologous approaches reduce donor–recipient alloimmune mismatch but require individualized tissue procurement, expansion, testing and product release. The optimized RESTART process involves approximately 14–20 days of post-thaw culture, excluding tissue procurement and quality-control activities, while cryopreservation of intermediate products provides scheduling flexibility[22]. Banked allogeneic products could shorten patient waiting times and facilitate larger-scale manufacture but introduce donor selection and banking requirements as well as potential alloimmune rejection. Comparative manufacturing costs, lead times and

clinical outcomes for human salivary organoid therapies remain unestablished[22],[N4]. Alloimmune rejection arising from donor–recipient incompatibility and recurrent autoimmunity in SjD represent distinct barriers: even autologous epithelial grafts may remain vulnerable to a persistently inflammatory host microenvironment[20],[139]. In autoimmune-prone NOD mice, human organoid transplantation improved salivary secretion at seven weeks without reducing lymphocytic focus scores[21], demonstrating short-term functional benefit despite ongoing inflammation but not long-term immune protection. Similarly, human organoid xenografts in immunodeficient recipients and murine organoid transplantation after irradiation do not establish long-term immune compatibility in human recipients[21],[77]. Potential strategies include donor–recipient compatibility assessment for allogeneic products and local or disease-directed immunomodulation in autoimmune recipients, although these approaches remain unvalidated for salivary organoid graft protection. Future studies should assess donor-cell persistence, graft-associated immune infiltration, epithelial injury and salivary function longitudinally in disease-relevant models.

Significant challenges remain: manufacturing is labor-intensive, demands rigorous GMP control and still depends in many protocols on complex matrices or open aseptic steps, raising scalability and regulatory concerns[22,77]. Whether regenerated epithelium can form functional drainage connections to the oral cavity is unresolved, as is whether transplanted cells achieve durable engraftment and sustained secretion in humans[21,22]. Early trial outcomes will therefore be decisive in defining therapeutic durability, safety, manufacturing feasibility and the regulatory pathway. Regenerative translation thus brings the three constraints together: sufficiently mature secretory cells, functional integration with the host gland and reproducible manufacture with established safety.

Chemical Reprogramming Culture and Epithelial Progenitor Cell Therapy

A central bottleneck for progenitor cell therapy is expanding human salivary epithelial cells to clinical scale: conventional 2D cultures senesce and cannot be maintained, while 3D organoids scale less efficiently and consume substantial matrix and growth factors. Chemical reprogramming culture (CRC) addresses this by using a defined small-molecule cocktail to expand human salivary basal progenitor cells (SG-BPCs) in feeder- and Matrigel-free 2D culture[79]. The strategy belongs to a broader class in which small molecules, rather than exogenous transcription factors, control human cell fate, and the same modality has been used to reprogram somatic cells to pluripotency and convert fibroblasts into functional epithelial cells for organ repair, establishing chemical reprogramming as a non-integrative, controllable route to therapeutic cells[170,171].

Mechanistically, CRC is grounded in niche-relevant signaling control. With the ROCK inhibitor Y-27632, the TGF-β inhibitor A83-01 and the BMP inhibitor LDN193189, dual TGF-β/BMP inhibition suppresses differentiation and sustains the basal progenitor state while ROCK inhibition limits senescence and apoptosis[79]. The same dual TGF-β/BMP inhibition module (A83-01 with LDN193189) forms the foundational cocktail for reprogramming fibroblasts into retinal pigment epithelium, underscoring that these modules transfer across epithelial systems[171]. Under these conditions, salivary epithelial cells surpassed 50 population doublings in roughly 80 days, exceeding the Hayflick limit that constrained cells grown without the molecules. High-passage karyotyping detected no gross chromosomal alterations in the cultures examined, although this does not exclude submicroscopic genomic changes or establish tumorigenic safety; the cocktail selectively enriched basal ductal progenitors retaining KRT5, KRT19 and SOX9 with a predominantly $CD49f^{+}/CD26^{-}$ phenotype rather than terminally differentiated luminal cells[79].

This 2D platform complements rather than replaces 3D organoid culture. The expanded SG-BPCs retained organoid-forming capacity and, when transferred to 3D culture, differentiated

toward acinar (AQP5, AMY1, BHLHA15) and myoepithelial (ACTA2) lineages, so organoid differentiation serves as a functional-validation step[79]. This defines a coherent pipeline: CRC provides large-scale progenitor expansion, organoid differentiation confirms lineage potential and function, and the expanded cells are applied as a product. Consistent with this, administration of SG-BPCs into a mouse model of radiation-induced hypofunction restored salivary function[79], mirroring outcomes reported for other chemically reprogrammed epithelial cells that integrate into host tissue and restore organ function[171].

This epithelial-progenitor approach is conceptually distinct from mesenchymal stromal cell (MSC) therapy. Whereas MSC paracrine effects are not tissue-specific and may incompletely target salivary epithelium, SG-BPCs offer tissue-specific differentiation with lineage-appropriate paracrine potential. Because CRC is chemically defined, feeder-free and non-integrative, it avoids exogenous transcription-factor delivery and a pluripotent intermediate, but these features alone do not establish tumorigenic safety. No dedicated in vivo teratoma formation or tumorigenicity assessment has been reported for these SG-BPCs, and the mouse efficacy study evaluated outcomes only up to eight weeks after cell administration[79]. Although teratoma testing primarily addresses the presence of pluripotent cells, expanded tissue-restricted progenitors still require appropriate assessment of abnormal or ectopic growth, passage-associated genetic alterations and long-term safety. Chemical reprogramming is therefore an attractive translational strategy, although clinical-scale manufacturing, potency testing and long-term safety remain to be established[170-172]. CRC-based epithelial progenitor therapy is best viewed as a potentially scalable, tissue-specific complement to organoid transplantation.

Extracellular Vesicle Strategies: Targeted Cell-Free Therapy

Beyond cell transplantation, extracellular vesicles (EVs) offer a cell-free strategy, with direct

salivary gland evidence from gland- or saliva-derived EV preparations evaluated in organoid assays and/or preclinical gland disease models[166,179,180]. More broadly, EVs from stem cells isolated from adipose tissue, urine and dental pulp, as well as from iPSCs, have shown anti-fibrotic and immunomodulatory activity in related models[173-176]. These products derive from non-salivary source cells and should be distinguished from salivary gland-derived EVs. CD9-enriched EVs from chemically reprogrammed basal progenitors exert anti-fibrotic effects in human salivary organoids and a mouse duct-obstruction model by delivering miR-3162 and miR-1290 to suppress ACVR1 signaling, with microfluidic chip-based purification and ductal delivery offering a minimally invasive route to the affected gland[166]. From an endogenous source, saliva-derived EVs attenuated TGF-β-induced fibrogenic activity in human salivary fibroblasts and organoids through STAT3 modulation[177], while EVs from salivary stem cells cultured on WNT3A-loaded microwell scaffolds conferred radioprotection in irradiated human salivary organoids and mouse glands via YWHAZ enrichment and PI3K–AKT activation in recipient cells, showing how biomaterial-defined niches tune EV cargo[178]. Unlike transplanted cells, EVs do not require persistent engraftment, but their clinical development still depends on product-specific potency, dose and manufacturing controls. Potency assays should measure biological activity relevant to the intended mechanism, such as attenuation of profibrotic signaling in salivary gland organoids or fibroblasts, rather than relying on EV identity markers or cargo abundance alone[166,177]. Dose reporting should include particle and protein measurements alongside functional activity, without treating any single measure as a universally validated dosing unit. Route-dependent gland retention, biodistribution, clearance and repeat-dose exposure require further evaluation. Manufacturing scale-up must ensure comparability across source-cell batches, EV isolation, formulation and storage while controlling non-vesicular contaminants and microbial impurities. Minimal Information for Studies of Extracellular Vesicles 2023 (MISEV2023) provides guidance for EV

characterization and functional reporting but does not establish clinical release specifications or replace GMP and product-specific potency validation[179] (Fig. 6).[179,180] (Fig. 6).

~~from gland- salivaapplications are derived from organoid- or CRC-cultured cells or validated using salivary gland organoid assay platforms[180,181].~~

FUTURE PERSPECTIVES AND EMERGING TECHNOLOGIES

Across the applications reviewed above, developmental maturity determines the range of physiological functions that organoids can reproduce, multicellular organization determines which tissue-level interactions they can capture, and standardization determines whether experimental advances can be translated into reliable clinical applications. These interdependent constraints motivate the following approaches to acinar differentiation, multicellular integration, bioengineering and clinical standardization.

Overcoming the Acinar Differentiation Barrier

A challenge that cuts across many applications of human salivary gland organoids is their incomplete and protocol-dependent differentiation toward mature acinar lineages[13,21,77]. Current strategies, including Notch inhibition, ALK/TGF-β modulation, FGF-based mesenchymal cues, and stromal or mesenchymal co-culture, have made incremental progress, but native-level human acinar maturation and durable secretory output are not yet consistently achieved across culture systems[13,18,21,89,182]. Future directions include identifying acinar-specific transcription-factor networks through single-cell multi-omics, developing temporally controlled protocols that sequentially activate progenitor expansion and acinar-commitment programs, and engineering culture environments that supply the mechanical and biochemical cues acinar maturation requires[2,21,36]. Incorporating cellular aging is a further consideration: age-associated changes such as senescent-cell accumulation and declining regenerative capacity are likely to influence organoid initiation, differentiation potential and regenerative performance, yet remain

insufficiently modeled[23,146,149].

Multicellular System Integration

The future of the field lies in more sophisticated models that incorporate the native tissue's neural, stromal, vascular, immune and extracellular-matrix compartments, which today's epithelial-only systems lack[36,135,147,182]. Direct experimental evidence for stromal and vascular regulation comes primarily from developmental models. In murine embryonic salivary organoids, PDGFRα⁺ stromal cells promoted proacinar differentiation through FGF2-responsive signaling involving BMP2 and BMP7[118]. In embryonic mouse salivary gland explants and reconstituted tissues, endothelial-cell depletion or VEGFR2 inhibition disrupted epithelial patterning and altered ductal and progenitor populations[119]. These findings demonstrate instructive roles for non-epithelial cells, although their contributions to mature human organoid secretion remain incompletely defined.

Neuronal integration presents both technical challenges and opportunities. Salivary glands are densely innervated by sympathetic and parasympathetic neurons that regulate secretion through complex neurotransmitter signaling[1], and incorporating peripheral neurons or neural components derived from organoids could enable studies of neuroepithelial interaction, autonomic regulation, neurogenic dysfunction and gland-specific pharmacology[36,47,162]. Proof-of-concept studies include engineered salivary-like constructs derived from human dental pulp stem cells, which exhibited neurostimulation-responsive calcium signaling and amylase production in vitro and stimulated epithelial and neuronal growth in an ex vivo damaged-gland model[90], as well as murine SGO–N2a co-cultures showing BoNT-responsive changes in secretory markers[162]. However, these findings do not establish stable, physiologically patterned autonomic innervation or long-term neurosecretory regulation following human organoid transplantation. The recent identification of TUBB3/VIM double-positive neural-like cells in chemically defined organoid cultures suggests that some conditions may support neural-like

stromal elements. If these prove to participate in neurosecretory signaling, they could reduce reliance on exogenous neuronal supplementation[21].

Vascular integration is essential for modeling drug exposure, hypoxic responses, immune trafficking and regeneration[21,147]. Perfused microphysiological systems and salivary tissue-chip platforms can provide controlled exposure, support high-content functional assays and improve modeling of radiation response and drug screening relative to static cultures[147], but faithful reconstruction of ductal flow, vascular perfusion and obstructive salivary disease physiology remains an unresolved engineering challenge[147,183].

Tumor microenvironment modeling with multicellular organoids is a compelling cancer application. Incorporating cancer-associated fibroblasts, immune cells, endothelial cells and matrix with tumor organoids may yield more accurate models of therapeutic responses, immune evasion and resistance[184,185], and patient-derived stromal and immune cells combined with tumor organoids could enable personalized microenvironment models that capture individual variation in tumor–stroma and tumor–immune interactions, although salivary gland cancer-specific implementations remain limited[16,101,113].

Bioengineering Solutions

3D bioprinting enables organoid or tissue-engineered structures with greater spatial control than self-organizing spheroids alone [27,186]. Precisely positioning epithelial, stromal, vascular and neural components could reconstruct branched acinar–ductal architecture and support more physiological cell–cell and cell–matrix interactions, but fabricating fully functional, transplantable constructs remains a long-term goal rather than a clinical reality[27,183]. Ultimately, combining bioprinting with organoid technology may enable implantable glandular constructs with defined architecture, vascular support and functional drainage, provided maturation, integration and safety barriers are overcome.

Beyond direct bioprinting, scaffold-guided spheroid engineering offers a complementary

strategy for controlling spheroid formation, phenotype, retrieval and delivery. A thermoresponsive fiber-based microwell scaffold supported uniform salivary gland stem-cell (SGSC) spheroid formation and enabled nonenzymatic retrieval by lowering the temperature[187]. Compared with monolayer cultures, scaffold-generated spheroids showed increased expression of stemness-associated genes, including LGR4, LGR5 and LGR6, and enhanced paracrine growth-factor expression[187]. Following implantation into irradiation-damaged mouse salivary glands, they reduced tissue injury, preserved acinar and ductal structures and promoted recovery of secretory function[187]. Building on this approach, a WNT3A-releasing double-layered microwell scaffold provided sustained biochemical stimulation during spheroid culture and increased EV production; the resulting EVs showed regenerative effects in irradiated human salivary organoids and a mouse model of radiation injury[178]. Together, these studies illustrate how engineered scaffolds can serve not only as structural platforms but also as bioactive systems for producing functionally optimized cell- and EV-based products. These findings remain preclinical and do not establish therapeutic efficacy in human recipients.

Automated culture systems and robotic platforms address scalability and standardization [123,188-190]. They can perform routine maintenance, media exchange, passaging, drug screening, imaging and analysis with greater consistency than manual workflows, while artificial intelligence and high-content imaging can assist quality control and phenotypic analysis[188,190]. For salivary gland organoids, such automation will be particularly important for reducing operator-dependent variability, shortening PDO screening timelines, and enabling GMP-compatible manufacturing of therapeutic cell products, as well as standardized production of scaffold-derived spheroids and EV-based products[22,189].

Standardization and Clinical Translation

The clinical translation of salivary gland organoid technologies requires standardization

frameworks that remain underdeveloped for this organ system [11,22,23]. Methodological variability among laboratories in tissue processing, media formulation, matrix composition, passaging, differentiation conditions and functional assessment criteria makes cross-study comparison of organoid phenotypes and drug sensitivity data difficult[16,22,77,123], so consensus protocols, standardized reference materials, shared reporting criteria and interlaboratory validation remain priorities for the reproducibility that scientific and regulatory communities require[11,123,179]. For precision oncology, this means defined benchmarks for tumor-cell authentication, minimum criteria for genomic and histological fidelity to parental tumors, and harmonized screening at clinically relevant concentrations, with turnaround shortened to align with clinical decision-making[16,101,113]. Research-grade standardization primarily supports reproducibility, whereas clinical-grade manufacture of therapeutic cell products additionally requires GMP-compliant processes, traceability and product-specific release criteria[22],[N4] (Table 6).

Regulatory pathways differ by application. Decision-support tests such as PDO drug sensitivity testing may fall within regulatory frameworks governing laboratory-developed tests (LDTs), in vitro diagnostic (IVD) tests or clinical laboratory services, with requirements varying by jurisdiction and intended use. Analytical performance, clinical validity and clinical utility should therefore be evaluated as distinct considerations[123]. In the United States, Clinical Laboratory Improvement Amendments (CLIA) requirements address laboratory quality but do not themselves establish clinical utility [N7]. The 2024 LDT rule issued by the U.S. Food and Drug Administration (FDA) was vacated in 2025, and the regulatory text was subsequently reverted [N8]. In the European Union, tests with an IVD intended purpose are subject to the In Vitro Diagnostic Medical Devices Regulation (IVDR), including the conditional health-institution exemption under Article 5(5) [N9].

Therapeutic applications involving organoid-derived cells require product- and

jurisdiction-specific oversight, which may include the European Union's advanced therapy medicinal product (ATMP) framework[22],[N4]. Product-specific quality and release criteria should address identity, purity, potency and sterility, whereas tumorigenicity and biodistribution are evaluated during product development, with risk-based long-term clinical follow-up [N4,N12]. For EV products, MISEV2023 provides a consensus framework for EV characterization and reporting but does not constitute a regulatory lot-release assay[179]. Functional pharmacology applications such as BoNT assays require product-specific validation and benchmarking against established potency assays[156,158,159]. Studies using murine salivary organoid–neuronal co-cultures have demonstrated BoNT-responsive changes in secretory markers[162], but these assays have not yet been incorporated into pharmacopoeial monographs.

Addressing current manufacturing limitations will require convergence of chemically defined matrices that replace xenogeneic substrates, closed or semi-closed automated culture that ensures batch consistency, and prospective registry studies capturing long-term functional outcomes[21,22,77]. Experience from the Dutch first-in-human program, South Korean preclinical platforms and the US salivary tissue-organoid biorepository will help shape practical manufacturing benchmarks and regulatory standards, although clinical efficacy still requires prospective demonstration[22,23,77,169].

CONCLUSION

Salivary gland organoid technology has reached a translational inflection point. What began as exploratory efforts to culture glandular tissue in three dimensions has matured into a field with early first-in-human regenerative translation[22,169], patient-derived tumor organoid libraries[16,17], functional pharmacology platforms[162], and emerging autoimmune and radiation-injury models[20,77,147].

Organoid generation from multiple cell sources, guided by defined signaling

dependencies across the WNT/R-spondin, FGF, EGF/NRG1, Notch and TGF-β/ALK pathways, has produced versatile tools for basic and translational work. Yet the persistent ductal- or basal/progenitor-dominant phenotype of many human systems remains a fundamental limitation for functional and regenerative applications; chemically defined culture, Notch and ALK/TGF-β modulation, and stromal cues have narrowed this gap without closing it[13,21]. Characterizing species-specific differences has clarified the complementary roles of each system, with murine organoids supporting studies of morphogenesis, secretory physiology, injury and development, and human organoids supporting patient-specific biology, precision medicine and translation[13,77].

The histological and molecular diversity of salivary gland tumors makes patient-derived organoids especially valuable, since organoids from rare subtypes that lack representative cell lines fill a critical preclinical gap. For non-neoplastic disease, patient-derived SjD organoids have helped characterize epithelial-intrinsic dysfunction and interferon/JAK–STAT activation, whereas future immune-integrated assembloids could extend this work to immune–epithelial crosstalk[20,135]. Existing organoids isolate epithelial programs and report drug responses such as JAK inhibition, while single-cell and spatial profiling identifies immune and stromal components that future chimeric models could incorporate[139]. These prospective extensions require direct validation in human salivary disease models before their contributions to pathogenesis can be established.

Functional pharmacology platforms extend organoids beyond disease modeling. Organoid-based BoNT assessment exemplifies this, offering a gland-specific neuroepithelial assay that complements established neuronal potency assays. Although salivary organoids are not yet suitable as regulatory release assays, their applicability to cholinergic agonist and secretagogue screening, neurosecretory pharmacodynamics and the evaluation of complex biologics such as EVs positions them as candidate platforms for future industrial and regulatory development under the 3R framework, subject to rigorous validation[156,162].

Regenerative applications are advancing along complementary routes. Clinical translation is emerging but unproven: the Dutch first-in-human program, South Korean collagen-based transplantation and United States biorepository efforts signal momentum rather than efficacy[22,23,77,169]. Chemical reprogramming culture is well suited to this pipeline because its feeder- and Matrigel-free, small-molecule-defined format supports prolonged expansion of tissue-specific progenitors that retain acinar and myoepithelial potential[79] and may facilitate scalable manufacturing. It may complement MSC- and iPSC-based approaches without requiring a pluripotent intermediate. Cell-free EV strategies may further broaden therapeutic options, although they remain largely preclinical[166].

Significant challenges remain: functional maturation beyond ductal- or basal/progenitor-dominant states, the scalability of GMP-compliant manufacturing that still relies on complex matrices and labor-intensive handling, the batch-to-batch variability of advanced multicellular models, and the need for long-term safety and efficacy data from early clinical programs. Addressing these interconnected developmental, architectural and translational challenges will require sustained collaboration among researchers, clinicians, regulators, industry and patient communities[21,22].

By analyzing species-specific organoid biology, tumor-subtype-specific precision oncology, autoimmune and radiation-injury modeling, functional pharmacology, regenerative transplantation and cell-free EV strategies in depth, this review has aimed to complement broader surveys of oral and maxillofacial organoids with the specificity that salivary gland challenges demand. How rapidly the technology reaches patients who currently have few effective options will depend on the outcomes of ongoing translational programs and on continued refinement of acinar differentiation, multicellular culture, standardized manufacturing and mechanism-specific cell-free therapies.

Data availability

No new datasets were generated or analyzed in this review.

Acknowledgments

The authors thank Dong-Su Jang, MFA, Medical Illustrator, Medical Research Support Section, Yonsei University College of Medicine, Seoul, Republic of Korea, for creating the illustrations and providing graphic support for this manuscript.

Funding

This work was supported by the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT (MSIT) (Grant No. RS-2023-00208601 and RS-2025-18362970), and by the Nano & Material Technology Development Program through the NRF funded by MSIT (Grant No. RS-2024-00449435).

Author contributions

Y.-J.Y. contributed to conceptualization, writing—original draft, visualization, reference curation, and writing—review and editing. Y.J.J. contributed to writing—original draft, reference curation, and writing—review and editing. M.J. contributed to visualization, reference curation, and writing—review and editing. S.H. contributed to writing—original draft, visualization, and reference curation. S.S. contributed to writing—original draft and reference curation. H.J. contributed to visualization. T.V.P. contributed to writing—original draft and reference curation. J.-Y.L. contributed to conceptualization, supervision, project administration, and writing—review and editing. All authors reviewed and approved the final manuscript.

Supplementary information

No supplementary information accompanies this article.

Competing Interests

The authors declare no competing interests.

Ethics approval

Not applicable. This article is a review and did not involve human participants or animals.

References

1 Proctor, G. B. & Carpenter, G. H. Regulation of salivary gland function by autonomic nerves. *Auton Neurosci* **133**, 3–18 (2007). https://doi.org/10.1016/j.autneu.2006.10.006

2 Hauser, B. R. *et al.* Generation of a Single-Cell RNAseq Atlas of Murine Salivary Gland Development. *iScience* **23**, 101838 (2020). https://doi.org/10.1016/j.isci.2020.101838

3 May, A. J. *et al.* Diverse progenitor cells preserve salivary gland ductal architecture after radiation-induced damage. *Development* **145** (2018). https://doi.org/10.1242/dev.166363

4 Proctor, G. B. & Carpenter, G. H. Salivary secretion: mechanism and neural regulation. *Monogr Oral Sci* **24**, 14–29 (2014). https://doi.org/10.1159/000358781

5 Hosseini, M. S. *et al.* Cancer treatment-related xerostomia: basics, therapeutics, and future perspectives. *Eur J Med Res* **29**, 571 (2024). https://doi.org/10.1186/s40001-024-02167-x

6 Talha, B. & Swarnkar, S. A. in *StatPearls* (2026).

7 Skalova, A., Hyrcza, M. D. & Leivo, I. Update from the 5th Edition of the World Health Organization Classification of Head and Neck Tumors: Salivary Glands. *Head Neck Pathol* **16**, 40–53 (2022). https://doi.org/10.1007/s12105-022-01420-1

8 Drost, J. & Clevers, H. Organoids in cancer research. *Nat Rev Cancer* **18**, 407–418 (2018). https://doi.org/10.1038/s41568-018-0007-6

9 Seo, Y. & Kim, H.-S. Emerging organoid-based platforms to study salivary gland hypofunction. *Organoid* **2**, e2 (2022). https://doi.org/10.51335/organoid.2022.2.e2

10 Clevers, H. Modeling Development and Disease with Organoids. *Cell* **165**, 1586–1597 (2016). https://doi.org/10.1016/j.cell.2016.05.082

11 Zhao, Z. *et al.* Organoids. *Nat Rev Methods Primers* **2** (2022). https://doi.org/10.1038/s43586-022-00174-y

12 Lombaert, I., Movahednia, M. M., Adine, C. & Ferreira, J. N. Concise Review: Salivary Gland Regeneration: Therapeutic Approaches from Stem Cells to Tissue Organoids. *Stem Cells* **35**, 97–105 (2017). https://doi.org/10.1002/stem.2455

13 Yoon, Y. J. *et al.* Salivary gland organoid culture maintains distinct glandular properties of murine and human major salivary glands. *Nat Commun* **13**, 3291 (2022). https://doi.org/10.1038/s41467-022-30934-z

14 Maimets, M. *et al.* Long-Term In Vitro Expansion of Salivary Gland Stem Cells Driven by Wnt Signals. *Stem Cell Reports* **6**, 150–162 (2016). https://doi.org/10.1016/j.stemcr.2015.11.009

15 Pringle, S. *et al.* Human Salivary Gland Stem Cells Functionally Restore Radiation Damaged Salivary Glands. *Stem Cells* **34**, 640–652 (2016). https://doi.org/10.1002/stem.2278

16 Lassche, G. *et al.* Development and characterization of patient-derived salivary gland cancer organoid cultures. *Oral Oncol* **135**, 106186 (2022). https://doi.org/10.1016/j.oraloncology.2022.106186

17 Wang, B. *et al.* An organoid library of salivary gland tumors reveals subtype-specific characteristics and biomarkers. *J Exp Clin Cancer Res* **41**, 350 (2022). https://doi.org/10.1186/s13046-022-02561-5

18 Moskwa, N. *et al.* Single-cell RNA sequencing reveals PDGFRalpha+ stromal cell subpopulations that promote proacinar cell differentiation in embryonic salivary gland organoids. *Development* **149** (2022). https://doi.org/10.1242/dev.200167

19 Kwon, H. R., Nelson, D. A., DeSantis, K. A., Morrissey, J. M. & Larsen, M. Endothelial cell regulation of salivary gland epithelial patterning. *Development* **144**, 211–220 (2017). https://doi.org/10.1242/dev.142497

20 Meudec, L. *et al.* Development of salivary gland organoids derived from patient biopsies: a functional model of Sjogren's disease. *Ann Rheum Dis* **84**, 1195–1206 (2025). https://doi.org/10.1016/j.ard.2025.04.020

21 Zhang, X. *et al.* Functional human salivary gland organoids for tissue regeneration in chemically defined culture systems. *Cell Rep Med* **7**, 102612 (2026). https://doi.org/10.1016/j.xcrm.2026.102612

22 Zanten, J. V. *et al.* Optimization of the Production Process of Clinical-Grade Human Salivary Gland Organoid-Derived Cell Therapy for the Treatment of Radiation-Induced Xerostomia in

Head and Neck Cancer. *Pharmaceutics* **16** (2024). https://doi.org/10.3390/pharmaceutics16030435
23 Aalam, S. M. M. *et al.* Establishment of salivary tissue-organoid biorepository: characterizing salivary gland stem/progenitor cells and novel differentiation marker PSMA/FOLH1. *NPJ Regen Med* **10**, 23 (2025). https://doi.org/10.1038/s41536-025-00410-5
24 Zhao, C. *et al.* Organoid Models for Salivary Gland Biology and Regenerative Medicine. *Stem Cells Int* **2021**, 9922597 (2021). https://doi.org/10.1155/2021/9922597
25 Li, S. *et al.* Organoids in Dentistry and Oral Medicine: From Disease Models to Regenerative Medicine. *Regenerative Medicine and Dentistry* **2**, 14 (2025). https://doi.org/10.53941/rmd.2025.100014
26 Pillai, S., Munguia-Lopez, J. G. & Tran, S. D. Bioengineered Salivary Gland MicrotissuesA Review of 3D Cellular Models and their Applications. *ACS Applied Bio Materials* **7**, 2620–2636 (2024). https://doi.org/10.1021/acsabm.4c00028
27 Klangprapan, J., Souza, G. R. & Ferreira, J. N. Bioprinting salivary gland models and their regenerative applications. *BDJ Open* **10**, 39 (2024). https://doi.org/10.1038/s41405-024-00219-2
28 Park, H. J., Cho, J., Oh, S. J. & Kim, H. S. Organoid-based modeling and regenerative strategies for salivary gland dysfunction. *Korean J Physiol Pharmacol* **29**, 683–698 (2025). https://doi.org/10.4196/kjpp.25.149
29 Barker, N. *et al.* Identification of stem cells in small intestine and colon by marker gene Lgr5. *Nature* **449**, 1003–1007 (2007). https://doi.org/10.1038/nature06196
30 Bullard, T. *et al.* Ascl3 expression marks a progenitor population of both acinar and ductal cells in mouse salivary glands. *Dev Biol* **320**, 72–78 (2008). https://doi.org/10.1016/j.ydbio.2008.04.018
31 Weng, P. L., Aure, M. H., Maruyama, T. & Ovitt, C. E. Limited Regeneration of Adult Salivary Glands after Severe Injury Involves Cellular Plasticity. *Cell Rep* **24**, 1464–1470 e1463 (2018). https://doi.org/10.1016/j.celrep.2018.07.016
32 Takamatsu, K. *et al.* Aging-associated stem/progenitor cell dysfunction in the salivary glands of mice. *Exp Cell Res* **409**, 112889 (2021). https://doi.org/10.1016/j.yexcr.2021.112889
33 Takahashi, K. *et al.* Induction of pluripotent stem cells from adult human fibroblasts by defined factors. *Cell* **131**, 861–872 (2007). https://doi.org/10.1016/j.cell.2007.11.019
34 Takahashi, K. & Yamanaka, S. Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors. *Cell* **126**, 663–676 (2006). https://doi.org/10.1016/j.cell.2006.07.024
35 Song, W. *et al.* Current developments and opportunities of pluripotent stem cells-based therapies for salivary gland hypofunction. *Front Cell Dev Biol* **12**, 1346996 (2024). https://doi.org/10.3389/fcell.2024.1346996
36 Tanaka, J. *et al.* Human induced pluripotent stem cell-derived salivary gland organoids model SARS-CoV-2 infection and replication. *Nat Cell Biol* **24**, 1595–1605 (2022). https://doi.org/10.1038/s41556-022-01007-6
37 Matsuno, E. *et al.* Human iPSC-derived salivary gland cell sheets integrate with injured glands to form glandular structures. *Stem Cell Reports* **20**, 102674 (2025). https://doi.org/10.1016/j.stemcr.2025.102674
38 Tanaka, J. *et al.* Generation of salivary glands derived from pluripotent stem cells via conditional blastocyst complementation. *Cell Rep* **43**, 114340 (2024). https://doi.org/10.1016/j.celrep.2024.114340
39 Heinzelmann, E. *et al.* iPSC-derived and Patient-Derived Organoids: Applications and challenges in scalability and reproducibility as pre-clinical models. *Curr Res Toxicol* **7**, 100197 (2024). https://doi.org/10.1016/j.crtox.2024.100197
40 Chang, M., Bogacheva, M. S. & Lou, Y. R. Challenges for the Applications of Human Pluripotent Stem Cell-Derived Liver Organoids. *Front Cell Dev Biol* **9**, 748576 (2021). https://doi.org/10.3389/fcell.2021.748576
41 Chatzeli, L., Gaete, M. & Tucker, A. S. Fgf10 and Sox9 are essential for the establishment of distal progenitor cells during mouse salivary gland development. *Development* **144**, 2294–2305 (2017). https://doi.org/10.1242/dev.146019

42 Emmerson, E. *et al.* SOX2 regulates acinar cell development in the salivary gland. *Elife* **6** (2017). https://doi.org/10.7554/eLife.26620

43 Farahat, M. *et al.* MSCs feeder layers induce SMG self-organization and branching morphogenesis. *PLoS One* **12**, e0176453 (2017). https://doi.org/10.1371/journal.pone.0176453

44 Hosseini, Z. F. *et al.* FGF2-dependent mesenchyme and laminin-111 are niche factors in salivary gland organoids. *J Cell Sci* **131** (2018). https://doi.org/10.1242/jcs.208728

45 MacDuffie, K. E. *et al.* Rescuing human fetal tissue research in the United States: A call for additional regulatory reform. *Stem Cell Reports* **16**, 2839–2843 (2021). https://doi.org/10.1016/j.stemcr.2021.10.016

46 Ehnes, D. D. *et al.* Sci-Seq of Human Fetal Salivary Tissue Introduces Human Transcriptional Paradigms and a Novel Cell Population. *Front Dent Med* **3** (2022). https://doi.org/10.3389/fdmed.2022.887057

47 Tanaka, J. *et al.* Generation of orthotopically functional salivary gland from embryonic stem cells. *Nat Commun* **9**, 4216 (2018). https://doi.org/10.1038/s41467-018-06469-7

48 Nusse, R. & Clevers, H. Wnt/beta-Catenin Signaling, Disease, and Emerging Therapeutic Modalities. *Cell* **169**, 985–999 (2017). https://doi.org/10.1016/j.cell.2017.05.016

49 Jarde, T. *et al.* Wnt and Neuregulin1/ErbB signalling extends 3D culture of hormone responsive mammary organoids. *Nat Commun* **7**, 13207 (2016). https://doi.org/10.1038/ncomms13207

50 de Lau, W. *et al.* Lgr5 homologues associate with Wnt receptors and mediate R-spondin signalling. *Nature* **476**, 293–297 (2011). https://doi.org/10.1038/nature10337

51 Yi, J. *et al.* Analysis of LGR4 receptor distribution in human and mouse tissues. *PLoS One* **8**, e78144 (2013). https://doi.org/10.1371/journal.pone.0078144

52 Verweij, L. H. G. *et al.* Tripotent Lgr5 stem cells in the posterior tongue generate lingual, taste, and salivary gland lineages. *Nat Commun* **16**, 10266 (2025). https://doi.org/10.1038/s41467-025-65140-0

53 de Lau, W., Peng, W. C., Gros, P. & Clevers, H. The R-spondin/Lgr5/Rnf43 module: regulator of Wnt signal strength. *Genes Dev* **28**, 305–316 (2014). https://doi.org/10.1101/gad.235473.113

54 Kim, K. A. *et al.* R-Spondin family members regulate the Wnt pathway by a common mechanism. *Mol Biol Cell* **19**, 2588–2596 (2008). https://doi.org/10.1091/mbc.e08-02-0187

55 Patel, V. N. & Hoffman, M. P. Salivary gland development: a template for regeneration. *Semin Cell Dev Biol* **25-26**, 52–60 (2014). https://doi.org/10.1016/j.semcdb.2013.12.001

56 Pasic, L. *et al.* Sustained activation of the HER1-ERK1/2-RSK signaling pathway controls myoepithelial cell fate in human mammary tissue. *Genes Dev* **25**, 1641–1653 (2011). https://doi.org/10.1101/gad.2025611

57 Zhang, Z. *et al.* Tumor Microenvironment-Derived NRG1 Promotes Antiandrogen Resistance in Prostate Cancer. *Cancer Cell* **38**, 279–296 e279 (2020). https://doi.org/10.1016/j.ccell.2020.06.005

58 Patel, N., Sharpe, P. T. & Miletich, I. Coordination of epithelial branching and salivary gland lumen formation by Wnt and FGF signals. *Dev Biol* **358**, 156–167 (2011). https://doi.org/10.1016/j.ydbio.2011.07.023

59 Steinberg, Z. *et al.* FGFR2b signaling regulates ex vivo submandibular gland epithelial cell proliferation and branching morphogenesis. *Development* **132**, 1223–1234 (2005). https://doi.org/10.1242/dev.01690

60 Ray, A. T. & Soriano, P. FGF signaling regulates salivary gland branching morphogenesis by modulating cell adhesion. *Development* **150** (2023). https://doi.org/10.1242/dev.201293

61 Kim, D., Yoon, Y. J., Choi, D., Kim, J. & Lim, J. Y. 3D Organoid Culture From Adult Salivary Gland Tissues as an ex vivo Modeling of Salivary Gland Morphogenesis. *Front Cell Dev Biol* **9**, 698292 (2021). https://doi.org/10.3389/fcell.2021.698292

62 Kang, B. K. *et al.* Maintenance of adult stem cells from human minor salivary glands via the Wnt signaling pathway. *Stem Cell Res Ther* **14**, 220 (2023). https://doi.org/10.1186/s13287-023-03445-x

63 Tojo, M. *et al.* The ALK-5 inhibitor A-83-01 inhibits Smad signaling and epithelial-to-mesenchymal transition by transforming growth factor-beta. *Cancer Sci* **96**, 791–800 (2005). https://doi.org/10.1111/j.1349-7006.2005.00103.x

64 Munoz Forti, K., Weisman, G. A. & Jasmer, K. J. Cell type-specific transforming growth factor-beta (TGF-beta) signaling in the regulation of salivary gland fibrosis and regeneration. *J Oral Biol Craniofac Res* **14**, 257–272 (2024). https://doi.org/10.1016/j.jobcr.2024.03.005
65 Zhang, X. *et al.* TGF-beta Pathway in Salivary Gland Fibrosis. *Int J Mol Sci* **21** (2020). https://doi.org/10.3390/ijms21239138
66 Dang, H. *et al.* Role for Notch signaling in salivary acinar cell growth and differentiation. *Dev Dyn* **238**, 724–731 (2009). https://doi.org/10.1002/dvdy.21875
67 Chatzeli, L. *et al.* A cellular hierarchy of Notch and Kras signaling controls cell fate specification in the developing mouse salivary gland. *Dev Cell* **58**, 94–109 e106 (2023). https://doi.org/10.1016/j.devcel.2022.12.009
68 Lin, C. *et al.* YAP is essential for mechanical force production and epithelial cell proliferation during lung branching morphogenesis. *Elife* **6** (2017). https://doi.org/10.7554/eLife.21130
69 Yu, F. X., Zhao, B. & Guan, K. L. Hippo Pathway in Organ Size Control, Tissue Homeostasis, and Cancer. *Cell* **163**, 811–828 (2015). https://doi.org/10.1016/j.cell.2015.10.044
70 Kozlowski, M. T., Crook, C. J. & Ku, H. T. Towards organoid culture without Matrigel. *Commun Biol* **4**, 1387 (2021). https://doi.org/10.1038/s42003-021-02910-8
71 Hughes, C. S., Postovit, L. M. & Lajoie, G. A. Matrigel: a complex protein mixture required for optimal growth of cell culture. *Proteomics* **10**, 1886–1890 (2010). https://doi.org/10.1002/pmic.200900758
72 Gjorevski, N. *et al.* Designer matrices for intestinal stem cell and organoid culture. *Nature* **539** 560–564 (2016). https://doi.org/10.1038/nature20168
73 Shubin, A. D. *et al.* Encapsulation of primary salivary gland cells in enzymatically degradable poly(ethylene glycol) hydrogels promotes acinar cell characteristics. *Acta Biomater* **50**, 437–449 (2017). https://doi.org/10.1016/j.actbio.2016.12.049
74 Wang, T. *et al.* Injectable decellularized extracellular matrix hydrogel promotes salivary gland regeneration via endogenous stem cell recruitment and suppression of fibrogenesis. *Acta Biomater* **169**, 256–272 (2023). https://doi.org/10.1016/j.actbio.2023.08.003
75 Munguia-Lopez, J. G. *et al.* Expansion of functional human salivary acinar cell spheroids with reversible thermo-ionically crosslinked 3D hydrogels. *Int J Oral Sci* **17**, 39 (2025). https://doi.org/10.1038/s41368-025-00368-6
76 Sato, T. *et al.* Long-term expansion of epithelial organoids from human colon, adenoma, adenocarcinoma, and Barrett's epithelium. *Gastroenterology* **141**, 1762–1772 (2011). https://doi.org/10.1053/j.gastro.2011.07.050
77 Jeon, S. G. *et al.* Salivary gland organoid transplantation as a therapeutic option for radiation-induced xerostomia. *Stem Cell Res Ther* **15**, 265 (2024). https://doi.org/10.1186/s13287-024-03833-x
78 Wollny, D. *et al.* Single-Cell Analysis Uncovers Clonal Acinar Cell Heterogeneity in the Adult Pancreas. *Dev Cell* **39**, 289–301 (2016). https://doi.org/10.1016/j.devcel.2016.10.002
79 Jeong, Y. J. *et al.* Chemical reprogramming culture for the expansion of salivary gland epithelial basal progenitor cells. *Stem Cell Res Ther* **16**, 187 (2025). https://doi.org/10.1186/s13287-025-04295-5
80 Cinat, D. *et al.* Notch signaling is a driver of glandular stem cell activity and regenerative migration after damage. *EMBO J* **45**, 374–393 (2026). https://doi.org/10.1038/s44318-025-00607-w
81 Krane, C. M. *et al.* Salivary acinar cells from aquaporin 5-deficient mice have decreased membrane water permeability and altered cell volume regulation. *J Biol Chem* **276**, 23413–23420 (2001). https://doi.org/10.1074/jbc.M008760200
82 Matsuzaki, T. *et al.* Function of the membrane water channel aquaporin-5 in the salivary gland. *Acta Histochem Cytochem* **45**, 251–259 (2012). https://doi.org/10.1267/ahc.12018
83 Nanduri, L. S. *et al.* Purification and ex vivo expansion of fully functional salivary gland stem cells. *Stem Cell Reports* **3**, 957–964 (2014). https://doi.org/10.1016/j.stemcr.2014.09.015
84 Kwak, M., Ninche, N., Klein, S., Saur, D. & Ghazizadeh, S. c-Kit(+) Cells in Adult Salivary Glands do not Function as Tissue Stem Cells. *Sci Rep* **8**, 14193 (2018). https://doi.org/10.1038/s41598-018-32557-1
85 Athwal, H. K. *et al.* Sox10 Regulates Plasticity of Epithelial Progenitors toward Secretory Units

of Exocrine Glands. *Stem Cell Reports* **12**, 366–380 (2019). https://doi.org/10.1016/j.stemcr.2019.01.002

86 Barker, N. *et al.* Lgr5(+ve) stem cells drive self-renewal in the stomach and build long-lived gastric units in vitro. *Cell Stem Cell* **6**, 25–36 (2010). https://doi.org/10.1016/j.stem.2009.11.013

87 Sato, T. *et al.* Single Lgr5 stem cells build crypt-villus structures in vitro without a mesenchymal niche. *Nature* **459**, 262–265 (2009). https://doi.org/10.1038/nature07935

88 Rocchi, C., Barazzuol, L. & Coppes, R. P. The evolving definition of salivary gland stem cells. *NPJ Regen Med* **6**, 4 (2021). https://doi.org/10.1038/s41536-020-00115-x

89 Yoshimoto, S. *et al.* Inhibition of Alk signaling promotes the induction of human salivary-gland-derived organoids. *Dis Model Mech* **13** (2020). https://doi.org/10.1242/dmm.045054

90 Adine, C., Ng, K. K., Rungarunlert, S., Souza, G. R. & Ferreira, J. N. Engineering innervated secretory epithelial organoids by magnetic three-dimensional bioprinting for stimulating epithelial growth in salivary glands. *Biomaterials* **180**, 52–66 (2018). https://doi.org/10.1016/j.biomaterials.2018.06.011

91 Dekkers, J. F. *et al.* A functional CFTR assay using primary cystic fibrosis intestinal organoids. *Nat Med* **19**, 939–945 (2013). https://doi.org/10.1038/nm.3201

92 Catalan, M. A. *et al.* A fluid secretion pathway unmasked by acinar-specific Tmem16A gene ablation in the adult mouse salivary gland. *Proc Natl Acad Sci U S A* **112**, 2263–2268 (2015). https://doi.org/10.1073/pnas.1415739112

93 Amano, O., Mizobe, K., Bando, Y. & Sakiyama, K. Anatomy and histology of rodent and human major salivary glands: -overview of the Japan salivary gland society-sponsored workshop. *Acta Histochem Cytochem* **45**, 241–250 (2012). https://doi.org/10.1267/ahc.12013

94 de Paula, F. *et al.* Overview of Human Salivary Glands: Highlights of Morphology and Developing Processes. *Anat Rec (Hoboken)* **300**, 1180–1188 (2017). https://doi.org/10.1002/ar.23569

95 Gao, X., Oei, M. S., Ovitt, C. E., Sincan, M. & Melvin, J. E. Transcriptional profiling reveals gland-specific differential expression in the three major salivary glands of the adult mouse. *Physiol Genomics* **50**, 263–271 (2018). https://doi.org/10.1152/physiolgenomics.00124.2017

96 Koslow, M., O'Keefe, K. J., Hosseini, Z. F., Nelson, D. A. & Larsen, M. ROCK inhibitor increases proacinar cells in adult salivary gland organoids. *Stem Cell Res* **41**, 101608 (2019). https://doi.org/10.1016/j.scr.2019.101608

97 Srinivasan, P. P. *et al.* Primary Salivary Human Stem/Progenitor Cells Undergo Microenvironment-Driven Acinar-Like Differentiation in Hyaluronate Hydrogel Culture. *Stem Cells Transl Med* **6**, 110–120 (2017). https://doi.org/10.5966/sctm.2016-0083

98 Shin, K. *et al.* Three-Dimensional Culture of Salivary Gland Stem Cell in Orthotropic Decellularized Extracellular Matrix Hydrogels. *Tissue Eng Part A* **25**, 1396–1403 (2019). https://doi.org/10.1089/ten.TEA.2018.0308

99 Saitou, M. *et al.* Functional Specialization of Human Salivary Glands and Origins of Proteins Intrinsic to Human Saliva. *Cell Rep* **33**, 108402 (2020). https://doi.org/10.1016/j.celrep.2020.108402

100 Skalova, A. & Hyrcza, M. D. Proceedings of the North American Society of Head and Neck Pathology Companion Meeting, New Orleans, LA, March 12, 2023: Classification of Salivary Gland Tumors: Remaining Controversial Issues? *Head Neck Pathol* **17**, 285–291 (2023). https://doi.org/10.1007/s12105-023-01541-1

101 Aizawa, Y. *et al.* Establishment of experimental salivary gland cancer models using organoid culture and patient-derived xenografting. *Cell Oncol (Dordr)* **46**, 409–421 (2023). https://doi.org/10.1007/s13402-022-00758-6

102 Lassche, G. *et al.* Precision oncology using organoids of a secretory carcinoma of the salivary gland treated with TRK-inhibitors. *Oral Oncol* **137**, 106297 (2023). https://doi.org/10.1016/j.oraloncology.2022.106297

103 Ding, H. *et al.* Current landscape and future trends in salivary gland oncology research-a bibliometric evaluation. *Gland Surg* **13**, 969–986 (2024). https://doi.org/10.21037/gs-24-94

104 Chen, J., Li, J. L., Chen, Z., Griffin, J. D. & Wu, L. Gene expression profiling analysis of CRTC1-MAML2 fusion oncogene-induced transcriptional program in human mucoepidermoid

carcinoma cells. *BMC Cancer* **15**, 803 (2015). https://doi.org/10.1186/s12885-015-1827-3
105 Chen, Z. *et al.* Aberrantly activated AREG-EGFR signaling is required for the growth and survival of CRTC1-MAML2 fusion-positive mucoepidermoid carcinoma cells. *Oncogene* **33**, 3869–3877 (2014). https://doi.org/10.1038/onc.2013.348
106 Chen, Z. *et al.* The CRTC1-MAML2 fusion is the major oncogenic driver in mucoepidermoid carcinoma. *JCI Insight* **6** (2021). https://doi.org/10.1172/jci.insight.139497
107 Persson, M. *et al.* Recurrent fusion of MYB and NFIB transcription factor genes in carcinomas of the breast and head and neck. *Proc Natl Acad Sci U S A* **106**, 18740–18744 (2009). https://doi.org/10.1073/pnas.0909114106
108 Mitani, Y. *et al.* Novel MYBL1 Gene Rearrangements with Recurrent MYBL1-NFIB Fusions in Salivary Adenoid Cystic Carcinomas Lacking t(6;9) Translocations. *Clin Cancer Res* **22**, 725–733 (2016). https://doi.org/10.1158/1078-0432.CCR-15-2867-T
109 Brayer, K. J., Frerich, C. A., Kang, H. & Ness, S. A. Recurrent Fusions in MYB and MYBL1 Define a Common, Transcription Factor-Driven Oncogenic Pathway in Salivary Gland Adenoid Cystic Carcinoma. *Cancer Discov* **6**, 176–187 (2016). https://doi.org/10.1158/2159-8290.CD-15-0859
110 Ho, A. S. *et al.* The mutational landscape of adenoid cystic carcinoma. *Nat Genet* **45**, 791–798 (2013). https://doi.org/10.1038/ng.2643
111 Urano, M. *et al.* Diagnostic Significance of HRAS Mutations in Epithelial-Myoepithelial Carcinomas Exhibiting a Broad Histopathologic Spectrum. *Am J Surg Pathol* **43**, 984–994 (2019). https://doi.org/10.1097/PAS.0000000000001258
112 Ferrarotto, R. *et al.* Proteogenomic Analysis of Salivary Adenoid Cystic Carcinomas Defines Molecular Subtypes and Identifies Therapeutic Targets. *Clin Cancer Res* **27**, 852–864 (2021). https://doi.org/10.1158/1078-0432.CCR-20-1192
113 Ishikawa, T. *et al.* Salivary gland cancer organoids are valid for preclinical genotype-oriented medical precision trials. *iScience* **26**, 106695 (2023). https://doi.org/10.1016/j.isci.2023.106695
114 Kondo, J. *et al.* Retaining cell–cell contact enables preparation and culture of spheroids composed of pure primary cancer cells from colorectal cancer. *Proceedings of the National Academy of Sciences* **108**, 6235–6240 (2011). https://doi.org/10.1073/pnas.1015938108
115 Tanaka, N. *et al.* Head and neck cancer organoids established by modification of the CTOS method can be used to predict in vivo drug sensitivity. *Oral Oncol* **87**, 49–57 (2018). https://doi.org/10.1016/j.oraloncology.2018.10.018
116 Wang, X. W. *et al.* Establishment of a patient-derived organoid model and living biobank for nasopharyngeal carcinoma. *Ann Transl Med* **10**, 526 (2022). https://doi.org/10.21037/atm-22-1076
117 de Kort, W. W. B. *et al.* Clinicopathological Factors as Predictors for Establishment of Patient Derived Head and Neck Squamous Cell Carcinoma Organoids. *Head Neck Pathol* **18**, 59 (2024). https://doi.org/10.1007/s12105-024-01658-x
118 Wallaschek, N. *et al.* Establishing Pure Cancer Organoid Cultures: Identification, Selection and Verification of Cancer Phenotypes and Genotypes. *J Mol Biol* **431**, 2884–2893 (2019). https://doi.org/10.1016/j.jmb.2019.05.031
119 Fisch, A. S. *et al.* Feasibility analysis of using patient-derived tumour organoids for treatment decision guidance in locally advanced head and neck squamous cell carcinoma. *Eur J Cancer* **213**, 115100 (2024). https://doi.org/10.1016/j.ejca.2024.115100
120 Iyer, J. *et al.* An Overview on the Histogenesis and Morphogenesis of Salivary Gland Neoplasms and Evolving Diagnostic Approaches. *Cancers (Basel)* **13** (2021). https://doi.org/10.3390/cancers13153910
121 Seethala, R. R. Basaloid/blue salivary gland tumors. *Modern Pathology* **30**, S84–S95 (2017). https://doi.org/10.1038/modpathol.2016.190
122 Vlachogiannis, G. *et al.* Patient-derived organoids model treatment response of metastatic gastrointestinal cancers. *Science* **359**, 920–926 (2018). https://doi.org/10.1126/science.aao2774
123 Driehuis, E., Kretzschmar, K. & Clevers, H. Establishment of patient-derived cancer organoids for drug-screening applications. *Nat Protoc* **15**, 3380–3409 (2020). https://doi.org/10.1038/s41596-020-0379-4
124 Geiger, J. L. *et al.* Management of Salivary Gland Malignancy: ASCO Guideline. *J Clin Oncol*

39, 1909–1941 (2021). https://doi.org/10.1200/JCO.21.00449
125 van Herpen, C. *et al.* Salivary gland cancer: ESMO-European Reference Network on Rare Adult Solid Cancers (EURACAN) Clinical Practice Guideline for diagnosis, treatment and follow-up. *ESMO Open* **7**, 100602 (2022). https://doi.org/10.1016/j.esmoop.2022.100602
126 Francies, H. E., Barthorpe, A., McLaren-Douglas, A., Barendt, W. J. & Garnett, M. J. Drug Sensitivity Assays of Human Cancer Organoid Cultures. *Methods Mol Biol* **1576**, 339–351 (2019). https://doi.org/10.1007/7651_2016_10
127 Veninga, V. & Voest, E. E. Tumor organoids: Opportunities and challenges to guide precision medicine. *Cancer Cell* **39**, 1190–1201 (2021). https://doi.org/10.1016/j.ccell.2021.07.020
128 Dalin, M. G. *et al.* Comprehensive Molecular Characterization of Salivary Duct Carcinoma Reveals Actionable Targets and Similarity to Apocrine Breast Cancer. *Clin Cancer Res* **22**, 4623–4633 (2016). https://doi.org/10.1158/1078-0432.CCR-16-0637
129 Millen, R. *et al.* Patient-derived head and neck cancer organoids allow treatment stratification and serve as a tool for biomarker validation and identification. *Med* **4**, 290–310 e212 (2023). https://doi.org/10.1016/j.medj.2023.04.003
130 Issing, C. *et al.* Head and neck tumor organoid biobank for modelling individual responses to radiation therapy according to the TP53/HPV status. *J Exp Clin Cancer Res* **44**, 85 (2025). https://doi.org/10.1186/s13046-025-03345-3
131 Wensink, G. E. *et al.* Patient-derived organoids as a predictive biomarker for treatment response in cancer patients. *npj Precision Oncology* **5**, 30 (2021). https://doi.org/10.1038/s41698-021-00168-1
132 Verstappen, G. M., Pringle, S., Bootsma, H. & Kroese, F. G. M. Epithelial-immune cell interplay in primary Sjogren syndrome salivary gland pathogenesis. *Nat Rev Rheumatol* **17**, 333–348 (2021). https://doi.org/10.1038/s41584-021-00605-2
133 Mariette, X. & Criswell, L. A. Primary Sjogren's Syndrome. *N Engl J Med* **378**, 931–939 (2018). https://doi.org/10.1056/NEJMcp1702514
134 Brito-Zeron, P. *et al.* Sjogren syndrome. *Nat Rev Dis Primers* **2**, 16047 (2016). https://doi.org/10.1038/nrdp.2016.47
135 Yang, T. *et al.* Salivary gland organoids and the future of modeling autoimmune epithelitis in Sjogren's disease. *Curr Opin Immunol* **98**, 102707 (2026). https://doi.org/10.1016/j.coi.2025.102707
136 Pringle, S. *et al.* Small-molecule inhibitors and the salivary gland epithelium in Sjogren's syndrome. *Expert Opin Investig Drugs* **28**, 605–616 (2019). https://doi.org/10.1080/13543784.2019.1631796
137 Lu, E. *et al.* Establishment of human minor salivary gland organoids in laminin-GelMA hydrogel from healthy individuals and Sjögren's disease patients. *Chemical Engineering Journal* **503**, 158257 (2025). https://doi.org/https://doi.org/10.1016/j.cej.2024.158257
138 Ewert, P. *et al.* Disruption of tight junction structure in salivary glands from Sjogren's syndrome patients is linked to proinflammatory cytokine exposure. *Arthritis Rheum* **62**, 1280–1289 (2010). https://doi.org/10.1002/art.27362
139 Inamo, J. *et al.* Comparative single-cell and spatial profiling of anti-SSA-positive and anti-centromere-positive Sjögren's disease reveals common and distinct immune activation and fibroblast-mediated inflammation. *Nature Communications* **16**, 8299 (2025). https://doi.org/10.1038/s41467-025-63935-9
140 Santos, A. J. M. *et al.* A human autoimmune organoid model reveals IL-7 function in coeliac disease. *Nature* **632**, 401–410 (2024). https://doi.org/10.1038/s41586-024-07716-2
141 Recaldin, T. *et al.* Human organoids with an autologous tissue-resident immune compartment. *Nature* **633**, 165–173 (2024). https://doi.org/10.1038/s41586-024-07791-5
142 Suhito, I. R., Sunil, C. & Tay, A. Engineering human immune organoids for translational immunology. *Bioact Mater* **44**, 164–183 (2025). https://doi.org/10.1016/j.bioactmat.2024.10.010
143 Onesto, M. M., Kim, J. I. & Pasca, S. P. Assembloid models of cell-cell interaction to study tissue and disease biology. *Cell Stem Cell* **31**, 1563–1573 (2024). https://doi.org/10.1016/j.stem.2024.09.017
144 Jasmer, K. J., Gilman, K. E., Munoz Forti, K., Weisman, G. A. & Limesand, K. H. Radiation-

Induced Salivary Gland Dysfunction: Mechanisms, Therapeutics and Future Directions. *J Clin Med* **9** (2020). https://doi.org/10.3390/jcm9124095

145 Mercadante, V. *et al.* Salivary Gland Hypofunction and/or Xerostomia Induced by Nonsurgical Cancer Therapies: ISOO/MASCC/ASCO Guideline. *J Clin Oncol* **39**, 2825–2843 (2021). https://doi.org/10.1200/JCO.21.01208

146 Peng, X. *et al.* Cellular senescence contributes to radiation-induced hyposalivation by affecting the stem/progenitor cell niche. *Cell Death Dis* **11**, 854 (2020). https://doi.org/10.1038/s41419-020-03074-9

147 Song, Y. *et al.* Development of a functional salivary gland tissue chip with potential for high-content drug screening. *Commun Biol* **4**, 361 (2021). https://doi.org/10.1038/s42003-021-01876-x

148 Nagle, P. W. *et al.* Lack of DNA Damage Response at Low Radiation Doses in Adult Stem Cells Contributes to Organ Dysfunction. *Clin Cancer Res* **24**, 6583–6593 (2018). https://doi.org/10.1158/1078-0432.CCR-18-0533

149 Marmary, Y. *et al.* Radiation-Induced Loss of Salivary Gland Function Is Driven by Cellular Senescence and Prevented by IL6 Modulation. *Cancer Res* **76**, 1170–1180 (2016). https://doi.org/10.1158/0008-5472.CAN-15-1671

150 van Luijk, P. *et al.* Sparing the region of the salivary gland containing stem cells preserves saliva production after radiotherapy for head and neck cancer. *Sci Transl Med* **7**, 305ra147 (2015). https://doi.org/10.1126/scitranslmed.aac4441

151 Nakamura, T. *et al.* M(3) muscarinic acetylcholine receptor plays a critical role in parasympathetic control of salivation in mice. *J Physiol* **558**, 561–575 (2004). https://doi.org/10.1113/jphysiol.2004.064626

152 Khalafalla, M. G. *et al.* P2 Receptors as Therapeutic Targets in the Salivary Gland: From Physiology to Dysfunction. *Front Pharmacol* **11**, 222 (2020). https://doi.org/10.3389/fphar.2020.00222

153 Adler, S. *et al.* The current scientific and legal status of alternative methods to the LD50 test for botulinum neurotoxin potency testing. The report and recommendations of a ZEBET Expert Meeting. *Altern Lab Anim* **38**, 315–330 (2010). https://doi.org/10.1177/026119291003800401

154 Bitz, S. The botulinum neurotoxin LD50 test - problems and solutions. *ALTEX* **27**, 114–116 (2010). https://doi.org/10.14573/altex.2010.2.114

155 Sesardic, D. & Das, R. Alternatives to the LD50 assay for botulinum toxin potency testing: Strategies and progress towards refinement, reduction and replacement. *AATEX* **14** (2008).

156 Fernandez-Salas, E. *et al.* Botulinum neurotoxin serotype A specific cell-based potency assay to replace the mouse bioassay. *PLoS One* **7**, e49516 (2012). https://doi.org/10.1371/journal.pone.0049516

157 Pellett, S. Progress in cell based assays for botulinum neurotoxin detection. *Curr Top Microbiol Immunol* **364**, 257–285 (2013). https://doi.org/10.1007/978-3-642-33570-9_12

158 Fonfria, E. *et al.* Replacement of the Mouse LD(50) Assay for Determination of the Potency of AbobotulinumtoxinA with a Cell-Based Method in Both Powder and Liquid Formulations. *Toxins (Basel)* **15** (2023). https://doi.org/10.3390/toxins15050314

159 Yang, Y., Zhang, H., Yuan, L., Wang, S. & Ma, X. A Validation of the Equivalence of the Cell-Based Potency Assay Method with a Mouse LD(50) Bioassay for the Potency Testing of OnabotulinumtoxinA. *Toxins (Basel)* **16** (2024). https://doi.org/10.3390/toxins16060279

160 Jost, W. H. *et al.* Botulinum neurotoxin type A in the interdisciplinary treatment of sialorrhea in adults and children-update and practice recommendations. *Front Neurol* **14**, 1275807 (2023). https://doi.org/10.3389/fneur.2023.1275807

161 Yang, C. L. *et al.* The effectiveness and safety of botulinum toxin injections for the treatment of sialorrhea with Parkinson's disease: a systematic review and meta-analysis. *BMC Pharmacol Toxicol* **24**, 52 (2023). https://doi.org/10.1186/s40360-023-00694-7

162 Yoon, Y. J. & Lim, J. Y. The Usefulness of Salivary Gland Organoids for Evaluation of the Potency of Botulinum Neurotoxin. *Laryngoscope* **134**, 2697–2704 (2024). https://doi.org/10.1002/lary.31312

163 Belaid, M. *et al.* Oral Delivery of Mesenchymal Stem Cell-Derived Extracellular Vesicles To Treat Intestinal Inflammation. *ACS Applied Materials & Interfaces* **18**, 38483–38501 (2026).

https://doi.org/10.1021/acsami.6c07496
164 Lu, B. *et al.* Therapeutic potential of mesenchymal stem cell-derived apoptotic vesicles in liver fibrosis: targeting the IGFBP3/PI3K-AKT axis via miR-409-3p. *Stem Cell Research & Therapy* (2026). https://doi.org/10.1186/s13287-026-05143-w
165 Yuan, X. L. *et al.* MSC-EVs attenuate subretinal fibrosis in choroidal neovascularization through miR-21-5p-mediated inhibition of EMT and MMT and suppression of inflammation. *J Neuroinflammation* **23** (2026). https://doi.org/10.1186/s12974-026-03836-w
166 Park, S. *et al.* CD9-enriched extracellular vesicles from chemically reprogrammed basal progenitors of salivary glands mitigate salivary gland fibrosis. *Bioact Mater* **47**, 229–247 (2025). https://doi.org/10.1016/j.bioactmat.2025.01.019
167 Kratochvil, M. J. *et al.* Engineered materials for organoid systems. *Nature Reviews Materials* **4** 606–622 (2019). https://doi.org/10.1038/s41578-019-0129-9
168 Yuki, K., Cheng, N., Nakano, M. & Kuo, C. J. Organoid Models of Tumor Immunology. *Trends in Immunology* **41**, 652–664 (2020). https://doi.org/https://doi.org/10.1016/j.it.2020.06.010
169 Verstegen, M. M. A. *et al.* Clinical applications of human organoids. *Nat Med* **31**, 409–421 (2025). https://doi.org/10.1038/s41591-024-03489-3
170 Guan, J. *et al.* Chemical reprogramming of human somatic cells to pluripotent stem cells. *Nature* **605**, 325–331 (2022). https://doi.org/10.1038/s41586-022-04593-5
171 Li, S. *et al.* Chemical reprogramming of fibroblasts into retinal pigment epithelium cells for vision restoration. *Nature Communications* **17**, 409 (2025). https://doi.org/10.1038/s41467-025-67104-w
172 Schoenfeldt, L. *et al.* Chemical reprogramming ameliorates cellular hallmarks of aging and extends lifespan. *EMBO Mol Med* **17**, 2071–2094 (2025). https://doi.org/10.1038/s44321-025-00265-9
173 Du, Z.-h. *et al.* SHED-Derived Exosomes Ameliorate Sjögren's Syndrome-Induced Hyposalivation by Suppressing Th1 Cell Response via the miR-29a-3p/T-bet Axis. *ACS Applied Materials & Interfaces* **17**, 5752–5761 (2025). https://doi.org/10.1021/acsami.4c16595
174 Guo, X. *et al.* Exosomes of human adipose stem cells mitigate irradiation injury to salivary glands by inhibiting epithelial-mesenchymal transition through miR-199a-3p targeting Twist1 and regulating TGFβ1/Smad3 pathway. *Theranostics* **15**, 1622–1641 (2025). https://doi.org/10.7150/thno.102346
175 Xiao, X. Y., Zhang, N. N., Long, Y. Z. & Huang, G. L. Repair mechanism of radiation-induced salivary gland injury by hypoxia-pretreated human urine-derived stem cell exosomes. *Oral Dis* **30**, 1234–1241 (2024). https://doi.org/10.1111/odi.14476
176 Kim, H. *et al.* Identification of Molecules Responsible for Therapeutic Effects of Extracellular Vesicles Produced from iPSC-Derived MSCs on Sjo¨gren’s Syndrome. *Aging and disease* **12**, 1409–1422 (2021). https://doi.org/10.14336/ad.2021.0621
177 Peng, B. *et al.* Saliva-derived extracellular vesicles: a promising therapeutic approach for salivary gland fibrosis. *J Transl Med* **23**, 593 (2025). https://doi.org/10.1186/s12967-025-06620-1
178 Cho, J. M. *et al.* Extracellular vesicles derived from salivary gland stem cells cultured on microwell scaffolds loaded with WNT3A promote the recovery of salivary gland function damaged by radiation via the YWHAZ-PI3K-AKT pathway. *Bioact Mater* **52**, 492–510 (2025). https://doi.org/10.1016/j.bioactmat.2025.06.024
179 Welsh, J. A. *et al.* Minimal information for studies of extracellular vesicles (MISEV2023): From basic to advanced approaches. *J Extracell Vesicles* **13**, e12404 (2024). https://doi.org/10.1002/jev2.12404
180 Kou, M. *et al.* Mesenchymal stem cell-derived extracellular vesicles for immunomodulation and regeneration: a next generation therapeutic tool? *Cell Death Dis* **13**, 580 (2022). https://doi.org/10.1038/s41419-022-05034-x
181 Sawutdeechaikul, P. *et al.* Mechanisms Tackling Salivary Gland Diseases with Extracellular Vesicle Therapies. *J Dent Res* **104**, 704–714 (2025). https://doi.org/10.1177/00220345251319295
182 Sui, Y. *et al.* Generation of functional salivary gland tissue from human submandibular gland

stem/progenitor cells. *Stem Cell Res Ther* **11**, 127 (2020). https://doi.org/10.1186/s13287-020-01628-4

183 Rose, S. C., Larsen, M., Xie, Y. & Sharfstein, S. T. Salivary Gland Bioengineering. *Bioengineering (Basel)* **11** (2023). https://doi.org/10.3390/bioengineering11010028

184 Gu, Z., Wu, Q., Shang, B., Zhang, K. & Zhang, W. Organoid co-culture models of the tumor microenvironment promote precision medicine. *Cancer Innov* **3**, e101 (2024). https://doi.org/10.1002/cai2.101

185 Thorel, L. *et al.* Patient-derived tumor organoids: a new avenue for preclinical research and precision medicine in oncology. *Exp Mol Med* **56**, 1531–1551 (2024). https://doi.org/10.1038/s12276-024-01272-5

186 Shopova, D., Yaneva, A., Mihaylova, A., Dinkova, A. & Bakova, D. Unlocking the Future: Bioprinting Salivary Glands-From Possibility to Reality. *J Funct Biomater* **15** (2024). https://doi.org/10.3390/jfb15060151

187 Hong, H. J. *et al.* Thermoresponsive fiber-based microwells capable of formation and retrieval of salivary gland stem cell spheroids for the regeneration of irradiation-damaged salivary glands. *J Tissue Eng* **13**, 20417314221085645 (2022). https://doi.org/10.1177/20417314221085645

188 Yang, H. *et al.* Bridging the organoid translational gap: integrating standardization and micropatterning for drug screening in clinical and pharmaceutical medicine. *Life Med* **3**, lnae016 (2024). https://doi.org/10.1093/lifemedi/lnae016

189 Kim, D. *et al.* From organoid culture to manufacturing: technologies for reproducible and scalable organoid production. *NPJ Biomed Innov* **3** (2026). https://doi.org/10.1038/s44385-025-00054-6

190 Zuo, J., Fang, Y., Wang, R. & Liang, S. High-throughput solutions in tumor organoids: from culture to drug screening. *Stem Cells* **43** (2025). https://doi.org/10.1093/stmcls/sxae070

Figure legends

Figure 1. Schematic representation of salivary gland anatomy and cellular organization.

The illustration depicts the detailed architecture of a salivary gland lobule, highlighting the relationship between acinar units (serous and mucous), the ductal system, and supporting cell types. (Inset) Anatomical locations of the major salivary glands: parotid, submandibular, and sublingual glands. Key structures include acinar cells (secretory units), myoepithelial cells (contractile function), and the sequential ductal system comprising intercalated, striated, and excretory ducts. Black arrows indicate the direction of saliva flow from acini through the ductal system to the oral cavity.

Figure 2. Network of signaling pathways regulating organoid development.

The diagram illustrates the major molecular signals involved in organoid expansion, maintenance, branching, and lineage differentiation. (1) Wnt/β-catenin signaling promotes proliferation through WNT3A or the GSK3β inhibitor CHIR99021. (2) R-spondin 1 potentiates Wnt signaling through LGR4, LGR5, and LGR6. (3) FGF7 and FGF10 activate FGFR2b-dependent ERK and AKT signaling to promote branching morphogenesis and counterbalance Wnt activity. (4) EGF-family ligands, including EGF, HB-EGF, and NRG1, activate MAPK and PI3K–AKT signaling to support organoid growth and survival. (5) Inhibition of BMP signaling by Noggin or dorsomorphin and inhibition of TGF-β signaling by A83-01 or SB431542 help preserve epithelial identity and limit EMT and fibrosis. (6) Notch signaling contributes to epithelial lineage regulation. DAPT inhibits NICD formation and is used in differentiation protocols that can promote acinar-marker expression.

Figure 3. Schematic representation of functional validation assays.

This multi-panel figure illustrates key methods for assessing organoid physiology. (a) Calcium

imaging: monitoring intracellular Ca^{2+} dynamics using fluorescent indicators such as Fluo-4 or GCaMP after representative carbachol stimulation. (b) Swelling assay: quantification of changes in organoid area as a measure of fluid secretion in response to acetylcholine or carbachol, norepinephrine or isoproterenol, vasoactive intestinal peptide, and forskolin. (c) Amylase secretion: measurement of amylase activity in collected culture medium using an enzymatic assay following secretagogue stimulation. (d) Chloride transport: conceptual assessment of CFTR- and ANO1/TMEM16A-mediated chloride flux using halide-sensitive fluorescent indicators, with forskolin and carbachol as representative stimuli.

Figure 4. Representative morphologies and molecular characteristics of species-specific organoid models.

(a) Under selected differentiation conditions, murine organoids can display robust branching morphogenesis and grape-like acinar clusters with expression of AQP5, AMY1, and MIST1, consistent with acinar differentiation and secretory-lineage maturation. (b) Human organoids commonly exhibit ductal-dominant morphologies, including solid dense, cystic, and simple differentiated structures. They show basal KRT5+/KRT14+ and luminal KRT7+/KRT19+ organization, with limited and protocol-dependent expression of acinar markers such as AQP5 and AMY1. (c) Strategies to enhance human organoid maturation include Notch inhibition with DAPT, TGF-β pathway modulation, and fibroblast co-culture, which may increase AQP5 and MIST1 expression and promote acinar differentiation. (d) Summary of major differences between murine and human organoids in morphology, acinar differentiation, stem/progenitor plasticity, and media or extracellular-matrix requirements.

Figure 5. Overview of salivary gland tumor diversity and the role of PDOs in capturing histological heterogeneity. The schematic depicts fusion-driven tumors, including

mucoepidermoid carcinoma (MEC) with CRTC1/3–MAML2 fusions, adenoid cystic carcinoma (ACC) with MYB/MYBL1–NFIB fusions, and secretory carcinoma with ETV6–NTRK3 fusions. Salivary duct carcinoma (SDC) is characterized by PI3K–AKT–mTOR pathway alterations, androgen receptor expression, and/or HER2 amplification. Rarer subtypes include epithelial–myoepithelial carcinoma (EMC) with HRAS mutations, carcinoma ex pleomorphic adenoma with PLAG1 or HRAS alterations, and polymorphous adenocarcinoma with PRKD1 alterations. Compared with conventional cell lines, patient-derived organoids can retain key histological and molecular features of parental tumors and provide patient-specific platforms for therapeutic-response testing, including TRK inhibitor evaluation in ETV6–NTRK3-positive secretory carcinoma.

Figure 6. Extracellular vesicle-based therapeutic strategies for salivary gland dysfunction.

The upper panel summarizes extracellular vesicles derived from diverse stem-cell sources, including adipose-derived stem cells, iPSC-derived mesenchymal stem cells, urine-derived stem cells, and stem cells from human exfoliated deciduous teeth. Representative therapeutic mechanisms include miR-199a-3p-mediated suppression of Twist1, TGF-β1, and Smad3, modulation of WNT3A/GSK3β signaling, and the miR-29a-3p/T-bet axis, which collectively target radiation injury, fibrosis, and immune dysregulation. The lower-left panel illustrates microfluidic isolation and ductal administration of CD9-positive EVs carrying miR-3162 and miR-1290, which suppress ACVR1 signaling and reduce fibrosis. The lower-middle panel shows saliva-derived small extracellular vesicles that attenuate fibroblast activation and fibrosis through STAT3 modulation. The lower-right panel depicts EVs derived from salivary gland stem cells cultured in WNT3A-loaded three-dimensional microwell scaffolds; YWHAZ-enriched EVs activate PI3K–AKT signaling in recipient salivary gland cells and promote

regeneration. EV, extracellular vesicle; iPSC, induced pluripotent stem cell; MSC, mesenchymal stem cell; SHED, stem cells from human exfoliated deciduous teeth; sEV, small extracellular vesicle